RATIONALITY OF THE TRIVIAL LATTICE RANK WEIGHTED MOTIVIC HEIGHT ZETA FUNCTION FOR ELLIPTIC SURFACES

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ABSTRACT. Let k be a perfect field with $\mathrm{char}(k) \neq 2, 3$, set $K = k(t)$, and let $\mathcal{W}_n^{\min}$ be the moduli stack of minimal elliptic curves over K of Faltings height n from the height–moduli framework of [BPS22] applied to $\overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4,6)$. For $[E] \in \mathcal{W}_n^{\min}$, let $S \to \mathbb{P}^1_k$ be the associated elliptic surface with section. Motivated by the Shioda–Tate formula, we consider the trivariate motivic height zeta function

$$\mathcal{Z}(u, v; t) := \sum_{n \geq 0} \Big(\sum_{[E] \in \mathcal{W}_n^{\min}} u^{T(S)} v^{\mathrm{rk}(E/K)} \Big) t^n$$

refining the height series by the trivial lattice rank $T(S)$ and Mordell–Weil rank $\mathrm{rk}(E/K)$. Setting $u = v = 1$ recovers the exact Euler product of [BPS22]. We prove that the trivial lattice specialization $Z_{\mathrm{Triv}}(u; t) = \mathcal{Z}(u, 1; t)$ is approximable to arbitrary dimensional precision by rational functions of $s = t^{1/12}$ in the dimensionally completed Grothendieck ring, combining exact Euler products on the isotrivial loci with a motivic discriminant stabilization adapting Vakil–Wood [VW15] to $\Delta = 4a_4^3 + 27a_6^2$. Over $k = \mathbb{C}$, the Hodge–Deligne realization of each $[u^m]Z_{\mathrm{Triv}}$ is strictly rational in s, with no completion required. We conjecture irrationality for the Néron–Severi $Z_{\mathrm{NS}}(w; t) = \mathcal{Z}(w, w; t)$ and Mordell–Weil $Z_{\mathrm{MW}}(v; t) = \mathcal{Z}(1, v; t)$ specializations. Applying the framework to all genus-0 modular curves, we show the torsion-weighted motivic height zeta function $Z_{\mathrm{tor}}(\tau; t)$ is a polynomial of degree at most 12 in τ with coefficients in $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}](t)$.

1. INTRODUCTION

Let k be a perfect field with $\mathrm{char}(k) \neq 2, 3$, and set $K := k(t)$. An elliptic curve E/K determines a relatively minimal elliptic surface with section

$$f : S \longrightarrow \mathbb{P}^1_k$$

unique up to isomorphism (see [Mir89, SS10] for background on elliptic surfaces).

The arithmetic of E/K is reflected in the geometry of S, and a basic organizing principle is the Shioda–Tate formula [Shi90]

$$\rho(S) \;=\; T(S) \;+\; \mathrm{rk}(E/K), \tag{1}$$

where $\rho(S) = \mathrm{rk}\,\mathrm{NS}(S_{\bar{k}})$ is the *geometric Picard rank*, $T(S)$ is the *rank of the geometric trivial lattice* generated by the zero section, a fiber class, and the components of reducible fibers not meeting the zero section, and $\mathrm{rk}(E/K)$ is the *Mordell–Weil rank*. For the relatively minimal elliptic surfaces $f : S \to \mathbb{P}^1_k$ with section considered in this paper, we have $q(S) = 0$ and $p_g(S) = n - 1$, hence the standard bounds

$$2 \leq \rho(S) \leq 10n, \qquad 2 \leq T(S) \leq 10n, \qquad 0 \leq \mathrm{rk}(E/K) \leq 10n - 2, \tag{2}$$

where $\rho(S) \leq 10n = h^{1,1}(S)$ is the Lefschetz bound over $k = \mathbb{C}$ (or in general Igusa's inequality $\rho(S) \leq b_2(S) = 12n - 2$).

In [BPS22], Bejleri–Park–Satriano construct height-moduli stacks of rational points on proper polarized cyclotomic stacks. In the fundamental modular curve example

$$\overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4,6), \qquad \lambda \simeq \mathcal{O}_{\mathcal{P}(4,6)}(1),$$

a minimal elliptic curve over K can be viewed as a rational point of λ–height n on $\overline{\mathcal{M}}_{1,1}$ over K. This yields a separated Deligne–Mumford stack of finite type

$$\mathcal{W}_n^{\min} := \mathcal{W}_{n,\mathbb{P}^1_k}^{\min}\big(\mathcal{P}(4,6), \mathcal{O}(1)\big)$$

parametrizing minimal elliptic curves over K of discriminant degree $12n$. Here a K-rational point of $\overline{\mathcal{M}}_{1,1}$ of λ-height n means the stacky height n with respect to the Hodge line bundle λ, in the sense of [ESZB23]. Under the identification $\overline{\mathcal{M}}_{1,1} \cong \mathcal{P}(4,6)$ one has $\lambda \simeq \mathcal{O}_{\mathcal{P}(4,6)}(1)$, and this height agrees with the Faltings height of the corresponding elliptic curve by [BPS22, Cor. 7.6].

Guided by (1), we introduce the following motivic generating series (see [Eke25] for background on the Grothendieck ring of stacks) refining the height generating series in [BPS22, §8] by weighting each height stratum with the *lattice ranks* of the associated relatively minimal elliptic surface.

Definition 1.1. Let k be a perfect field of characteristic $\neq 2, 3$, and consider the height–moduli stack

$$\mathcal{W}_n^{\min} = \mathcal{W}_{n,\mathbb{P}^1_k}^{\min}\big(\mathcal{P}(4,6), \mathcal{O}(1)\big)$$

parametrizing minimal elliptic curves over $K = k(t)$ of discriminant height $12n$. The *trivariate height zeta function* is

$$\mathcal{Z}(u,v;t) := \sum_{n\geq 0}\left(\sum_{[E]\in\mathcal{W}_n^{\min}} u^{T(S)}\, v^{\mathrm{rk}(E/K)}\right) t^n$$

where for each $[E] \in \mathcal{W}_n^{\min}$ we write $S \to \mathbb{P}^1_k$ for the associated relatively minimal elliptic surface $f : S \to \mathbb{P}^1_k$ with section, and:

- $T(S)$ is the rank of the trivial lattice of S;
- $\mathrm{rk}(E/K)$ is the Mordell–Weil rank.

The following specializations are the associated *bivariate height zeta functions*:

$$Z_{\mathrm{Triv}}(u;t) := \mathcal{Z}(u,1;t), \tag{3}$$

$$Z_{\mathrm{MW}}(v;t) := \mathcal{Z}(1,v;t), \tag{4}$$

$$Z_{\mathrm{NS}}(w;t) := \mathcal{Z}(w,w;t). \tag{5}$$

The specialization $Z_{\mathrm{Triv}}(u;t) = \mathcal{Z}(u,1;t)$ is rigorously defined in $K_0(\mathrm{Stck}_k)[u][\![t]\!]$, since $T(S)$ is constant on each Kodaira stratum. The specializations $Z_{\mathrm{MW}}(v;t)$ and $Z_{\mathrm{NS}}(w;t)$ are well-defined as weighted point counting series over $\mathbb{F}_q$ but not motivically in general: for a given Picard rank ρ (or Mordell–Weil rank r), there are infinitely many Néron–Severi lattice types realizing that rank, each contributing a separate Noether–Lefschetz stratum [CDK95], so the rank-ρ locus is a countable — but not finite constructible — union. We therefore work motivically in u throughout and treat the v- and w-gradings as point counting refinements.

Setting $u = v = 1$ forgets the lattice rank grading and specializes to the *univariate motivic height zeta function* $Z_{\vec{\lambda}}(t) = \mathcal{Z}(1,1;t) \in K_0(\mathrm{Stck}_k)[\![t]\!]$ and likewise to

its inertial refinement $\mathcal{I}Z_{\bar{\lambda}}(t)$ which encodes the totality of rational points on $\overline{\mathcal{M}}_{1,1}$ over $K = k(t)$. [BPS22, Thm. 8.9] shows that both series are in fact rational in t, i.e. lie in $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}](t)$, and gives explicit formulas.

The assumption $\mathrm{char}(k) \neq 2, 3$ is used throughout in two essential ways: first, it ensures the existence of the short Weierstrass form $y^2 = x^3 + a_4 x + a_6$ (equivalently, the isomorphism $\overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4,6)$ over $\mathbb{Z}\left[\frac{1}{6}\right]$); second, it guarantees that the Kodaira–Néron fiber classification [Kod63, Né64] and Tate correspondence (i.e. *Tate's algorithm* [Tat75] *via twisted maps* [BPS22, Thm. 7.12]) apply in their standard form.

In this paper we focus on $Z_{\mathrm{Triv}}(u;t)$. The key point is that the trivial lattice is governed by *local bad reduction*: its rank is determined by the geometric Kodaira fiber configuration of $\pi_{\bar{k}} \colon S_{\bar{k}} \to \mathbb{P}^1_{\bar{k}}$. Writing $\mathrm{Triv}(S) \subset \mathrm{NS}(S_{\bar{k}})$ for the geometric trivial lattice and $T(S) := \mathrm{rk}(\mathrm{Triv}(S))$, we have the following explicit formula.

Lemma 1.2. *Let $\pi \colon S \to \mathbb{P}^1_k$ be a relatively minimal elliptic surface with section, and let $\mathfrak{f}$ be the multiset of singular fibers of $\pi_{\bar{k}} \colon S_{\bar{k}} \to \mathbb{P}^1_{\bar{k}}$. If m_v denotes the number of irreducible components of the fiber at v, then*

$$T(S) \;=\; 2 + \sum_{v \in \mathfrak{f}} (m_v - 1).$$

Definition 1.3. Fix $n \geq 1$. For a geometric fiber configuration $\mathfrak{f}$, let $\mathcal{W}_n^{\min,(\mathfrak{f})} \subset \mathcal{W}_n^{\min}$ denote the locus parametrizing those $[E] \in \mathcal{W}_n^{\min}$ whose associated surface $S_{\bar{k}} \to \mathbb{P}^1_{\bar{k}}$ has singular fiber configuration $\mathfrak{f}$ (cf. [BPS22, Thm. 7.16]).

Definition 1.4. Fix $n \geq 0$. By Proposition 2.1, $\mathcal{W}_n^{\min}$ admits a *finite* constructible stratification by Kodaira data, and $T(S)$ is constant on each stratum. For $n \geq 1$ and each T with $2 \leq T \leq 10n$, let

$$\mathcal{W}_n^{\min}(T) \subset \mathcal{W}_n^{\min}$$

be the finite union of those Kodaira strata on which $T(S) = T$ (hence a finite union of locally closed substacks). For $n = 0$, set $\mathcal{W}_0^{\min} := \mathcal{W}_0^{\min}(2)$.

The trivial–lattice–rank–weighted motivic height zeta function is

$$Z_{\mathrm{Triv}}(u;t) := \sum_{n \geq 0} \sum_{T \geq 2} u^T \left\{ \mathcal{W}_n^{\min}(T) \right\} t^n \;\in\; K_0(\mathrm{Stck}_k)[u][\![t]\!].$$

We prove that $Z_{\mathrm{Triv}}(u;t)$ is approximately rational in $s = t^{1/12}$ in the dimensionally completed Grothendieck ring, reflecting the local nature of the trivial lattice.

Theorem 1.5. *Let k be a perfect field with $\mathrm{char}(k) \neq 2, 3$, put $s = t^{1/12}$, and let $\widehat{\mathcal{M}}_{\mathbb{L}}$ denote the completion of $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}]$ with respect to the dimensional filtration* [Kon95, VW15]. *Then for each $m \geq 2$ and each $N \geq 0$, there exists a rational function $R_{m,N}(s) \in \widehat{\mathcal{M}}_{\mathbb{L}}(s)$ such that for all n sufficiently large (depending on m and N), the normalized error*

$$\left([u^m] Z_{\mathrm{Triv}}(u;t) \big|_{t^n} - R_{m,N}(s) \big|_{s^{12n}} \right) \cdot \mathbb{L}^{-10n}$$

has dimension $\leq -(N+1)$ in $\widehat{\mathcal{M}}_{\mathbb{L}}$. Equivalently,

$$[u^m] Z_{\mathrm{Triv}}(u;t) \;\in\; \overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)} \;\subset\; \widehat{\mathcal{M}}_{\mathbb{L}}[\![s]\!]$$

for each m, where the closure is taken with respect to the coefficient-wise dimensional filtration topology.

The approximating rational functions arise from a three-locus partition of the moduli (non-constant j-invariant, $j \equiv 0$, and $j \equiv 1728$):

(i) *On the isotrivial loci $j \equiv 0$ and $j \equiv 1728$, the discriminant is a pure power of a single Weierstrass coefficient, all local conditions are linear, and the resulting Euler products are exact at all heights (Proposition 2.8). These are* strictly *rational: they lie in $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}](s)$ itself, without completion or approximation.*

(ii) *On the non-constant-j locus, the additive fiber types (types* II *through* II**) are handled by linear independence of vanishing conditions on the Weierstrass coefficients, yielding an Euler product that is exact above an explicit height threshold (Lemma 2.7, Proposition 2.9).*

(iii) *The multiplicative residual (root multiplicities of $\Delta = 4a_4^3 + 27a_6^2$ at coprime points) is handled by a motivic discriminant stabilization adapting the method of Vakil–Wood* [VW15]*: a truncated inclusion–exclusion at depth N produces the rational approximation $R_{m,N}$, with error vanishing in the dimensional filtration as $N \to \infty$ (Proposition 2.11).*

Setting $u = 1$ recovers the exact Euler product of [BPS22, Thm. 8.9]*: the approximation becomes exact, and strict rationality holds. See Remark 2.15 for the extension to $K = k(C)$ for a general smooth projective curve C. See Theorem 2.12 for the proof.*

Over $k = \mathbb{C}$, the Hodge–Deligne motivic measure $\mathrm{HD}\colon K_0(\mathrm{Stck}_{\mathbb{C}})[\mathbb{L}^{-1}] \to \mathbb{Z}[x, y, (xy)^{-1}]$, $\mathbb{L} \mapsto xy$, converts approximate motivic rationality into strict rationality, via the weight bound in mixed Hodge theory [Eke25].

Theorem 1.6. *Let $k = \mathbb{C}$, $K = \mathbb{C}(t)$, and $s = t^{1/12}$. Then for each $m \geq 2$, the generating function*

$$\sum_{n \geq 0} \mathrm{HD}\big(\{\mathcal{W}_n^{\min}(m)\} \cdot \mathbb{L}^{-10n}\big) s^{12n} \in \mathbb{Z}[x, y, (xy)^{-1}](s)$$

is a rational function of s, with no completion required. More precisely, there exists a Laurent polynomial $K_m \in \mathbb{Z}[x, y, (xy)^{-1}]$ such that for each bidegree $(p_0, q_0) \in \mathbb{Z}^2$ and for all n sufficiently large (depending on m and (p_0, q_0)),

$$e_c^{p_0, q_0}\big(\{\mathcal{W}_n^{\min}(m)\} \cdot \mathbb{L}^{-10n}\big) = [x^{p_0} y^{q_0}] K_m.$$

That is, each virtual Hodge number of $\{\mathcal{W}_n^{\min}(m)\} \cdot \mathbb{L}^{-10n}$ stabilizes at a finite height threshold. See Theorem 2.13 for the proof.

The proof combines a motivic local-to-global factorization in the style of Kapranov [Kap00, CLL16] with the twisted-map stratification of the height–moduli $\mathcal{W}_n^{\min}$ from [BPS22] and the evaluation morphisms of [BM96]. Throughout we use the Bejleri–Park–Satriano correspondence [BPS22, Thm. 3.3] between rational points, minimal weighted linear series, and twisted morphisms; in particular, local reduction conditions are encoded by representable twisted morphisms to $\overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4, 6)$, yielding a moduli-theoretic Tate correspondence compatible with the minimal model program. Unordered collections of local factors supported at distinct points of $\mathbb{P}^1$ are governed by symmetric powers $\mathrm{Sym}^N(\mathbb{P}^1)$. We reorganize these

symmetric-power contributions using the power structure on $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}]$, and we record the resulting identity in Lemma 2.7.

The additive local factors (types II through II*) impose linear vanishing conditions on the Weierstrass coefficients (a_4, a_6), and the sparsity argument of Lemma 2.7 shows these are independent at large heights. On the isotrivial loci $j \equiv 0$ and $j \equiv 1728$—the same loci that produce the lower order main terms in [BPS22, Thm. 9.7]—the discriminant reduces to a pure power of a single Weierstrass coefficient, and all local conditions become linear. These loci admit their own exact Euler products (Proposition 2.8), with no correction at any height.

The multiplicative sector (cusp families I_k and I_k^*) presents a fundamental obstruction: the discriminant $\Delta = 4a_4^3 + 27a_6^2$ is a *nonlinear* function of the Weierstrass data, and the source space $H^0(\mathcal{O}(4n)) \oplus H^0(\mathcal{O}(6n))$ (dimension $10n + 2$) is too small to independently prescribe discriminant root multiplicities at all singular points (the total discriminant degree is $12n$). The motivic transversality of Lemma 2.6 handles the regime where the multiplicative sector carries at most $\frac{1}{3}$ of the discriminant degree, but the generic semistable elliptic surface lies outside this range. To resolve this, we adapt the motivic stabilization method of Vakil–Wood [VW15]: for each fixed trivial lattice rank m, there are finitely many "visible" fiber configurations (those contributing positively to m), and the residual simple roots of Δ are handled by a truncated inclusion–exclusion that converges in the dimensional completion $\widehat{\mathcal{M}}_{\mathbb{L}}$ (Proposition 2.11). At $u = 1$ the trivial lattice grading is forgotten, the visible/invisible distinction collapses, and both the small-height additive corrections and the discriminant stabilization errors vanish identically, recovering the exact Euler product of [BPS22, Thm. 8.9].

2. Rationality of the trivial lattice rank weighted case

Throughout, let k be a perfect field with $\mathrm{char}(k) \neq 2, 3$, set $K = k(t)$, and let $\pi : S \to \mathbb{P}^1_k$ be the relatively minimal elliptic surface with section associated to E/K. Write $\mathrm{Triv}(S) \subset \mathrm{NS}(S_{\bar{k}})$ for the geometric trivial lattice and $T(S) = \mathrm{rk}\,\mathrm{Triv}(S)$.

Proposition 2.1. *Fix $n \geq 1$. The discriminant degree constraint $\sum_v e(F_v) = 12n$ implies that only finitely many geometric fiber configurations $\mathfrak{f}$ occur among surfaces parametrized by $\mathcal{W}_n^{\min}$. Consequently,*

$$\mathcal{W}_n^{\min} = \bigsqcup_{\mathfrak{f}} \mathcal{W}_n^{\min,(\mathfrak{f})}$$

is a finite constructible stratification. Moreover, the trivial lattice rank $T(S)$ is constant on each stratum $\mathcal{W}_n^{\min,(\mathfrak{f})}$.

Proof. The Kodaira–Néron classification gives $e(\mathrm{I}_k) = k$, $e(\mathrm{I}_k^*) = k + 6$, and $e(F_v) \in \{2, 3, 4, 6, 8, 9, 10\}$ for the remaining types. Since $\sum_v e(F_v) = 12n$, only finitely many multisets of Kodaira symbols can occur, yielding the finite stratification by [BPS22, Thm. 7.16]. Constancy of $T(S)$ on each stratum follows from Lemma 1.2. ■

A multivariate height series. We briefly recall the local indexing used in the twisted-maps description of height–moduli. By [BPS22, Thm. 5.1] the height-n moduli stack $\mathcal{M}_{n,C}(\mathcal{X},\mathcal{L})$ on a proper polarized cyclotomic stack $\mathcal{X}$ with polarizing line bundle $\mathcal{L}$ admits a finite stratification by locally closed substacks indexed by admissible local conditions and degrees: there is a finite disjoint union of morphisms

$$\bigsqcup_{\Gamma,d} \mathcal{H}^{\Gamma}_{d,C}(\mathcal{X},\mathcal{L})/S_\Gamma \longrightarrow \mathcal{M}_{n,C}(\mathcal{X},\mathcal{L}),$$

where $\mathcal{H}^{\Gamma}_{d,C}(\mathcal{X},\mathcal{L})$ is the moduli stack of representable twisted morphisms of stable height d to $(\mathcal{X},\mathcal{L})$ with local twisting conditions

$$\Gamma = \big(\{r_1,a_1\},\dots,\{r_s,a_s\}\big),$$

recording the stabilizer orders r_i and the corresponding characters a_i at the stacky marked points of the source root stack. The indices (Γ,d) range over those satisfying the height decomposition formula

$$n \;=\; d \;+\; \sum_{i=1}^{s} \frac{a_i}{r_i}.$$

Here $S_\Gamma \subset S_s$ is the subgroup permuting stacky marked points of the same local type.

Definition 2.2. For the Euler-product argument it is useful to distinguish *local factor types* from *evaluation labels*. Let $\mathcal{I}\overline{\mathcal{M}}_{1,1} \simeq \mathcal{IP}(4,6)$ be the inertia stack of $\overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4,6)$ over $\mathbb{Z}\big[\frac{1}{6}\big]$ with $\mathcal{L} = \mathcal{O}(1)$ the Hodge line bundle (see [HP23, §2] for background on inertia stacks).

(1) Local factor types. Let J denote the finite set of local factor types occurring in the Tate algorithm stratification via twisted maps (see [BPS22, §7]); concretely one may take

$$J = \Big\{\mathrm{II},\ \mathrm{III},\ \mathrm{IV},\ \mathrm{II}^*,\ \mathrm{III}^*,\ \mathrm{IV}^*,\ \mathrm{I}_0^*(j\neq 0,1728),\ \mathrm{I}_0^*(j=0),\ \mathrm{I}_0^*(j=1728),\ \mathrm{I}_\bullet,\ \mathrm{I}_\bullet^*\Big\},$$

where $\mathrm{I}_\bullet$ and $\mathrm{I}_\bullet^*$ are the two cusp *shapes* over $j=\infty$.

(2) Evaluation labels. Let $\mathcal{A}$ denote the set of evaluation labels used to index evaluation conditions, i.e. the inertia components in which the evaluation maps land. Away from the cusp $j=\infty$, the inertia label determines the Kodaira symbol, so the non-cusp labels form a finite set

$$\mathcal{A}_{\mathrm{nc}} = \Big\{\mathrm{II},\ \mathrm{III},\ \mathrm{IV},\ \mathrm{II}^*,\ \mathrm{III}^*,\ \mathrm{IV}^*,\ \mathrm{I}_0^*(j\neq 0,1728),\ \mathrm{I}_0^*(j=0),\ \mathrm{I}_0^*(j=1728)\Big\}.$$

At the cusp $j=\infty$, the inertia label records only the cusp shape ($\mathrm{I}_\bullet$ or $\mathrm{I}_\bullet^*$); the additional integer $k\geq 1$ (contact order with the boundary, equivalently the pole order of j) is *not* part of the local twisting conditions Γ of [BPS22, Def. 3.2]: for I_k one has $(r,a)=(0,0)$ so no stacky marking appears, while for I_k^* one has $(r,a)=(2,1)$ independently of k. The contact order is instead determined by the discriminant valuation of the Weierstrass model, equivalently the pole order of the j-map at $j=\infty$ ([BPS22, Thm. 7.12]). In the generating function $\mathcal{H}(s;\mathbf{x})$, the

contact order becomes a free summation variable, collapsed by geometric resummation (Lemma 2.4). Accordingly we set

$$\mathcal{A} \coloneqq \mathcal{A}_{\mathrm{nc}} \sqcup \{\mathrm{I}_\bullet,\ \mathrm{I}_\bullet^*\}.$$

For $\alpha \in \mathcal{A}_{\mathrm{nc}}$, let $m(\alpha) \in \mathbb{Z}_{\geq 1}$ be the number of irreducible components of the corresponding Kodaira fiber, so that $m(\alpha)-1$ is its contribution to the trivial lattice. For the cusp shapes $\mathrm{I}_\bullet$ and $\mathrm{I}_\bullet^*$, the component number depends on the contact order $k \geq 1$; this k-dependence is incorporated by geometric resummation (Lemma 2.4). Note that the same Kodaira symbol may correspond to distinct inertia components; for example, I_0^* splits according to whether $j \in \{0, 1728\}$ or $j \notin \{0, 1728\}$.

Definition 2.3. Fix an auxiliary variable s with $s^{12} = t$. Introduce variables $\{x_\alpha\}_{\alpha\in\mathcal{A}}$ and define

$$\mathcal{H}(s;\mathbf{x}) \coloneqq \sum_{n\geq 0} \sum_{\mathfrak{f}} \Big(\prod_{v\in\mathfrak{f}} x_{\alpha_v}\Big)\big\{\mathcal{W}_n^{\min,(\mathfrak{f})}\big\} s^{12n} \in K_0(\mathrm{Stck}_k)[\mathbf{x}][\![s]\!], \tag{6}$$

where for fixed n the inner sum ranges over the finitely many geometric fiber configurations $\mathfrak{f}$ occurring in height n (Proposition 2.1).

For each singular fiber F_v in $\mathfrak{f}$, let $\alpha_v \in \mathcal{A}$ denote the corresponding inertia/evaluation label. Away from the cusp $j = \infty$ this label is the Kodaira symbol, while over $j = \infty$ it records only the cusp shape $\mathrm{I}_\bullet$ or $\mathrm{I}_\bullet^*$. The additional contact order $k \geq 1$ at the cusp is part of the twisted-maps chart data which is *not* recorded by the variables x_α.

Kapranov zeta function. For a k–variety (or Deligne–Mumford stack) X, we write

$$\zeta_X(y) \coloneqq \sum_{N\geq 0} \big\{\mathrm{Sym}^N(X)\big\}\, y^N \in K_0(\mathrm{Stck}_k)[\![y]\!]$$

for the Kapranov motivic zeta function [Kap00]. For $X = \mathbb{P}^1$ one has $\zeta_{\mathbb{P}^1}(y) = 1/\big((1-y)(1-\mathbb{L}\,y)\big)$.

Cusp resummation and motivic transversality.

Lemma 2.4. *Let R be a commutative ring.*

(1) Geometric resummation *Fix $A \in R$. For integers $a, c \geq 1$ and $b, d \geq 0$, one has in $R[\![u,t]\!]$*

$$\sum_{k\geq 1} A u^{ak+b} t^{ck+d} = A u^{a+b} t^{c+d} \cdot \frac{1}{1-u^a t^c}. \tag{7}$$

Moreover, if $k_1, \ldots, k_M \geq 1$ are independent and contribute multiplicatively with the same step (a, c), then

$$\sum_{k_1,\ldots,k_M\geq 1} A \prod_{i=1}^{M} u^{ak_i+b} t^{ck_i+d} = A\Big(u^{a+b} t^{c+d}\Big)^M \cdot \frac{1}{(1-u^a t^c)^M}. \tag{8}$$

Equivalently, each marking contributes one factor $(1-u^a t^c)^{-1}$, so M such markings contribute the power $(1-u^a t^c)^{-M}$, up to the monomial shift $\big(u^{a+b} t^{c+d}\big)^M$.

(2) Cusp shapes for Z_{Triv} *Assume* $\mathrm{char}(k) \neq 2,3$ *and work in* $R = K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}]$. *Introduce an auxiliary variable* s *with* $t = s^{12}$, *so that* t^n *corresponds to* $\deg(\Delta) = 12n$, *while* s *records the integral discriminant degree* $\deg(\Delta)$.

After specializing $x_\beta = u^{m(\beta)-1}$ *for* $\beta \in \mathcal{A}_{\mathrm{nc}}$, *a cusp marking of shape* $\mathrm{I}_\bullet$ *(resp.* $\mathrm{I}^*_\bullet$*) with contact order* $k \geq 1$ *contributes weight* $u^{k-1}s^k$ *(resp.* $u^{k+4}s^{k+6}$*), since*

$$m(\mathrm{I}_k) - 1 = k - 1, \qquad \nu(\Delta) = k, \qquad m(\mathrm{I}^*_k) - 1 = k + 4, \qquad \nu(\Delta) = k + 6.$$

Hence summing over $k \geq 1$ *at a single cusp marking gives, in* $R[\![u, s]\!]$,

$$x_{\mathrm{I}_\bullet} = \sum_{k\geq 1} u^{k-1}s^k = \frac{s}{1-us}, \qquad x_{\mathrm{I}^*_\bullet} = \sum_{k\geq 1} u^{k+4}s^{k+6} = \frac{u^5 s^7}{1-us}. \tag{9}$$

In particular, each cusp marking of either shape contributes one factor $(1-us)^{-1}$ *after resummation. Thus a factor type* j *with* $\beta_{j,\mathrm{I}_\bullet}$ *markings of shape* $\mathrm{I}_\bullet$ *and* $\beta_{j,\mathrm{I}^*_\bullet}$ *markings of shape* $\mathrm{I}^*_\bullet$ *contributes the cusp factor*

$$(1-us)^{-(\beta_{j,\mathrm{I}_\bullet} + \beta_{j,\mathrm{I}^*_\bullet})},$$

together with the monomial shift

$$u^{5\beta_{j,\mathrm{I}^*_\bullet}} s^{\beta_{j,\mathrm{I}_\bullet} + 7\beta_{j,\mathrm{I}^*_\bullet}}$$

coming from (9).

Proof. For (7), factor out the $k = 1$ term and sum the geometric series. Equation (8) follows by independence of the k_i. Part (2) specializes to the cusp weights $u^{k-1}s^k$ and $u^{k+4}s^{k+6}$ via (7) with $(a, c) = (1, 1)$. ■

Definition 2.5. For each $\ell \geq 1$, the *local multiplicative locus* is

$$\Lambda(\ell) := \big\{(\alpha_0, \ldots, \alpha_\ell,\ \beta_0, \ldots, \beta_\ell) \in \mathbb{A}^{\ell+1} \times \mathbb{A}^{\ell+1} \;\big|\; \mathrm{ord}_0(4A^3 + 27B^2) = \ell,\ (\alpha_0, \beta_0) \neq (0,0)\big\},$$

where $A = \sum_{j=0}^{\ell} \alpha_j z^j$ and $B = \sum_{j=0}^{\ell} \beta_j z^j$. The class $\{\Lambda(\ell)\} \in K_0(\mathrm{Var}_k)$ depends only on ℓ, not on n or the choice of point.

The nonlinearity of $\Delta = 4a_4^3 + 27a_6^2$ means that prescribing discriminant root multiplicities at r points requires controlling k_i-jets of (a_4, a_6), not just values. The following lemma shows that the resulting jet conditions are independent whenever the total jet demand $\sum_i (k_i + 1)$ is at most $4n + 1$.

Lemma 2.6 (Motivic transversality)**.** *Let* k *be a perfect field with* $\mathrm{char}(k) \neq 2, 3$. *Fix* r *distinct closed points* $v_1, \ldots, v_r \in \mathbb{P}^1$ *and positive integers* $k_1, \ldots, k_r \geq 1$. *For* $n \geq 1$, *define the* multiplicative multi-point locus $\mathcal{M}_n(k_\bullet, v_\bullet) :=$

$$\big\{(a_4, a_6) \in H^0\big(\mathcal{O}(4n)\big) \oplus H^0\big(\mathcal{O}(6n)\big) \;\big|\; \mathrm{ord}_{v_i}(4a_4^3 + 27a_6^2) = k_i,\ (a_4(v_i), a_6(v_i)) \neq (0,0)\ \ \forall\, i\big\}$$

and the jet evaluation map

$$\mathrm{ev}\colon H^0\big(\mathcal{O}(4n)\big) \oplus H^0\big(\mathcal{O}(6n)\big) \longrightarrow \bigoplus_{i=1}^{r} \Big(J^{k_i}_{v_i}\big(\mathcal{O}(4n)\big) \oplus J^{k_i}_{v_i}\big(\mathcal{O}(6n)\big)\Big).$$

If $\sum_{i=1}^{r} (k_i + 1) \leq 4n + 1$, *then* ev *is surjective and*

$$\big\{\mathcal{M}_n(k_\bullet, v_\bullet)\big\} = \mathbb{L}^{10n+2-\sum_i 2(k_i+1)} \cdot \prod_{i=1}^{r} \big\{\Lambda(k_i)\big\}. \tag{10}$$

Proof. The condition $\mathrm{ord}_{v_i}(\Delta) = k_i$ depends only on the k_i-jet of (a_4, a_6) at v_i, since $\Delta^{(j)}(v_i)$ for $j \le k_i$ is determined by $\{a_4^{(\ell)}(v_i), a_6^{(\ell)}(v_i)\}_{\ell \le j}$ by the Leibniz rule. The jet evaluation map ev decomposes as the direct sum of Hermite interpolation maps for a_4 and a_6 separately; the former requires $\sum_i (k_i+1) \le 4n+1 = \dim H^0(\mathcal{O}(4n))$ conditions, the latter $\sum_i (k_i+1) \le 6n+1$, so ev is surjective under the stated bound. Since ev is a surjective linear map with kernel $\mathbb{A}^{10n+2-\sum_i 2(k_i+1)}$, the vector bundle relation gives $\{\mathrm{ev}^{-1}(Z)\} = \mathbb{L}^{\dim\ker} \cdot \{Z\}$ for any constructible Z in the target. The conditions at distinct points v_i act on disjoint jet factors, so $Z = \prod_i \Lambda(k_i)$ and (10) follows. ■

In the multiplicative sector, the discriminant budget gives $\sum_i k_i = 12n - D_{\mathrm{add}}$, so this bound fails whenever the multiplicative fibers carry more than roughly $\frac{1}{3}$ of the discriminant degree — in particular, for the generic semistable elliptic surface.

For a fixed trivial lattice rank m, the *visible* fibers (I_k with $k \ge 2$, I_k^* with $k \ge 1$, and the additive types with $m_v - 1 \ge 1$) have total jet demand $D(\sigma) \le D(m)$ bounded independently of n. By the lemma (for the multiplicative visible fibers) and the sparsity argument (Lemma 2.7, Step 3, for the additive visible fibers), the jet evaluation map at the σ-support points is surjective for $n \ge \lceil (D(m)-1)/4 \rceil$. The residual I_1 fibers (invisible, with $m_v - 1 = 0$) do not enter the jet evaluation map; their contribution is handled by the motivic discriminant stabilization (Proposition 2.11).

Euler product decomposition.

Lemma 2.7. *Let $\mathcal{A}$ be the finite set of evaluation labels from Definition 2.2, and let $\mathcal{H}(s;\mathbf{x})$ be the multivariate height series defined in* (6)*. After inverting $\mathbb{L}$, the series $\mathcal{H}(s;\mathbf{x})$ admits a motivic Euler product decomposition: for each monomial in the* additive *variables $\{x_\alpha\}_{\alpha \in \mathcal{A}_{\mathrm{nc}}}$, the coefficient is a rational function of s with coefficients in $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}]$ (as a formal power series in the cusp variables $x_{\mathrm{I}_\bullet}, x_{\mathrm{I}_\bullet^*}$).*

More precisely, there exist:

- *a finite index set J of local factor types,*
- *motivic classes $A_j \in K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}]$,*
- *integers $c_j \ge 0$, recording the discriminant degree increment contributed by one local factor of type j (with $c_j \ge 1$ for additive types and $c_j = 0$ for cusp types, whose s-dependence is deferred to the cusp variables),*
- *and exponents $\beta_{j,\alpha} \in \mathbb{Z}_{\ge 0}$ for $\alpha \in \mathcal{A}$, recording how many markings of inertia type α occur in a local factor of type j,*

such that, writing

$$Y_j(s;\mathbf{x}) := A_j \Big(\prod_{\alpha \in \mathcal{A}} x_\alpha^{\beta_{j,\alpha}} \Big) s^{c_j},$$

one has

$$\mathcal{H}(s;\mathbf{x}) = \prod_{j \in J} \big(1 - Y_j(s;\mathbf{x})\big)^{-\{\mathbb{P}^1\}} + P(s;\mathbf{x}), \tag{11}$$

where $P(s;\mathbf{x}) \in K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}][\![s,\mathbf{x}]\!]$ is a formal power series whose s-structure is resolved by the visible/invisible decomposition and motivic discriminant stabilization

(Proposition 2.11) in the Main Theorem (Theorem 2.12). Here the power-structure exponent $\{\mathbb{P}^1\}$ *means*

$$(1-Y_j)^{-\{\mathbb{P}^1\}} = \sum_{N\geq 0}\{\mathrm{Sym}^N(\mathbb{P}^1)\}\,Y_j^N = \frac{1}{(1-Y_j)(1-\mathbb{L}\,Y_j)}. \tag{12}$$

Moreover, for $\alpha \in \{\mathrm{I}_\bullet, \mathrm{I}_\bullet^*\}$ *the exponent* $\beta_{j,\alpha}$ *counts only the number of cusp markings of the given cusp* shape *in factor type* j*; it does not record the contact order* $k \geq 1$.

Proof. By [BPS22, Thm. 3.3 & Prop. 5.8], the correspondence between twisted maps $\mathcal{C} \to \mathcal{P}(4,6)$ and minimal weighted linear series on $\mathbb{P}^1_k$ yields, for each n, a finite locally closed stratification of $\mathcal{W}_n^{\min}$ into charts

$$\mathcal{H}^{\Gamma}_{d,\mathbb{P}^1}\big(\mathcal{P}(4,6),\mathcal{O}(1)\big)\big/S_\Gamma \;\cong\; \mathcal{R}^{\mu}_n/S_\Gamma$$

indexed by admissible local conditions $\Gamma = \big(\{r_1,a_1\},\dots,\{r_s,a_s\}\big)$ and stable height d, where $n = d + \sum a_i/r_i$ and the partition μ corresponds to Γ via the normalized base locus [BPS22, Prop. 3.9].

Local factor coefficients. Let J be the finite set of local factor types in Definition 2.2, each recording a pair (r_j, a_j) of local twisting data together with the j-invariant of the fiber (equivalently, a vanishing condition $\gamma_j = (\nu(a_4), \nu(a_6))$ as in [BPS22, Table of Thm. 1.6]). For each $j \in J$, consider the stratum $\mathcal{W}^{\gamma_j}_{n,\mathbb{P}^1}(4,6)$ of minimal Weierstrass fibrations over $\mathbb{P}^1$ with *exactly one* singular fiber of type j at a varying point and smooth or multiplicative fibers elsewhere. By [BPS22, Prop. 6.7], $\mathcal{W}^{\gamma_j}_n$ is a Zariski-locally trivial fibration over $\mathbb{P}^1$ with fiber $[R^{\gamma_j}_n(0)/\mathbb{G}_m]$, where $R^{\gamma_j}_n(0) \subset H^0(\mathcal{O}(4n)) \oplus H^0(\mathcal{O}(6n))$ is the linear subspace defined by the vanishing conditions $\nu_0(a_4) \geq p_j$, $\nu_0(a_6) \geq q_j$ at a fixed point $0 \in \mathbb{P}^1$. One has the equality

$$\big\{\mathcal{W}^{\gamma_j}_n\big\} \;=\; \big\{\mathbb{P}^1\big\}\cdot\frac{\big\{R^{\gamma_j}_n(0)\big\}}{\{\mathbb{G}_m\}} \tag{13}$$

in $K_0(\mathrm{Stck}_k)$. By [BPS22, Prop. 9.4], for $n \gg 0$ the motivic class is

$$\big\{\mathcal{W}^{\gamma_j}_n\big\} \;=\; (\mathbb{L}^2-1)\,\mathbb{L}^{10n-p_j-q_j}$$

where (p_j, q_j) are the Weierstrass vanishing orders from γ_j. We define the *local factor coefficient* A_j to be the normalized one-fiber motivic class

$$A_j \;:=\; \frac{\big\{R^{\gamma_j}_n(0)\big\}}{\{\mathbb{G}_m\}\cdot\mathbb{L}^{10n}} \;=\; \frac{\big\{\mathcal{W}^{\gamma_j}_n\big\}}{\{\mathbb{P}^1\}\cdot\mathbb{L}^{10n}} \;\in\; K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}].$$

This class is *independent of* n: from the explicit formula above we read off

$$A_j \;=\; \frac{(\mathbb{L}^2-1)\,\mathbb{L}^{-p_j-q_j}}{\mathbb{L}+1} \;=\; (\mathbb{L}-1)\,\mathbb{L}^{-p_j-q_j},$$

For the additive types $j \in J_{\mathrm{add}}$ this gives the complete list of local factor coefficients; more generally each entry in [BPS22, Table of Thm. 1.8] yields the corresponding A_j after dividing by $\{\mathbb{P}^1\}\cdot\mathbb{L}^{10n}$. For the multiplicative types $j = \mathrm{I}_\bullet$, the local factor coefficient $A_{\mathrm{I}_k} = \mathbb{L}^{18}$ for all $k \geq 1$ is computed separately in [HP19, Cor. 2] (see also [PS21]); the formula $A_j = (\mathbb{L}-1)\mathbb{L}^{-p_j-q_j}$ does not apply to this case since $(p_j, q_j) = (0,0)$ and the discriminant condition $\nu_x(\Delta) = k$ is nonlinear. For additive types $j \in J_{\mathrm{add}}$, the discriminant degree increment c_j is the s-degree

contributed by the local factor: explicitly, c_j equals the discriminant valuation of a fiber of type j. For cusp types $j \in \{\mathrm{I}_\bullet, \mathrm{I}^*_\bullet\}$, the discriminant valuation depends on the contact order k and is not recorded by c_j; instead we set $c_j = 0$ and defer the s-dependence to the cusp variables $x_{\mathrm{I}_\bullet}, x_{\mathrm{I}^*_\bullet}$, whose s-content is assigned by the cusp resummation (Lemma 2.4). The exponents $\beta_{j,\alpha}$ recording the evaluation labels are determined by γ_j via the Tate correspondence (i.e. *Tate's algorithm via twisted maps*) [BPS22, Thm. 7.12].

Euler product structure. By [BPS22, Thm. 2.30], the chart with s ordered stacky markings maps via the forgetful morphism

$$\mathcal{H}^{\Gamma}_{d,\mathbb{P}^1}\big(\mathcal{P}(4,6), \mathcal{O}(1)\big) \longrightarrow \mathrm{Conf}_s(\mathbb{P}^1)$$

recording the support points of the markings. Under the isomorphism $\mathcal{H}^{\Gamma}_d \cong \mathcal{R}^{\mu}_n$ of [BPS22, Prop. 5.8], this is identified with the projection $\mathcal{R}^{\mu}_n \to \mathrm{Conf}_{l(\mu)}(\mathbb{P}^1)$ sending a weighted linear series to the support of its normalized base locus.

The argument proceeds by decomposing the height series according to the *additive fiber configuration* and appealing to the Kapranov zeta function separately for each slice. We partition the set of local factor types as $J = J_{\mathrm{mult}} \sqcup J_{\mathrm{add}}$, where $J_{\mathrm{mult}} = \{\mathrm{I}_\bullet\}$ consists of the multiplicative types and J_{add} consists of the additive types (II, III, IV, $\mathrm{I}^*_0 (j \neq 0, 1728)$, $\mathrm{I}^*_0 (j \in \{0, 1728\})$, $\mathrm{I}^*_\bullet$, IV*, III*, II*). By Tate's algorithm [BPS22, Thm. 7.12], a multiplicative fiber of type I_k satisfies $\nu_x(a_4) = 0$ and $\nu_x(a_6) = 0$ (the Weierstrass coefficients are units at x); the contact order $k \geq 1$ is determined by $\nu_x(\Delta) = k$. By contrast, each additive type j imposes $\nu_x(a_4) \geq p_j \geq 1$ and $\nu_x(a_6) \geq q_j \geq 1$.

Step 1: Multiplicative fibers and the discriminant. Since multiplicative fibers of type I_k have $(p_j, q_j) = (0, 0)$, they impose no vanishing conditions on $a_4 \in H^0(\mathbb{P}^1, \mathcal{O}(4n))$ or $a_6 \in H^0(\mathbb{P}^1, \mathcal{O}(6n))$ and do not appear in the normalized base locus of the weighted linear series. Their positions and contact orders are instead *determined by* the discriminant $\Delta = 4a_4^3 + 27a_6^2$, which is a *nonlinear* function of the Weierstrass data. The local factor coefficient $A_{\mathrm{I}_k} = \mathbb{L}^{18}$ and the cusp variable conventions are as in the construction above; the passage from the single-point class to a multi-point Euler product requires inter-point independence of discriminant root multiplicities. The motivic transversality (Lemma 2.6) establishes this when $\sum_i (k_i + 1) \leq 4n + 1$, but this condition fails in the multiplicative-heavy regime. The full resolution uses the motivic discriminant stabilization (Proposition 2.11) in the proof of Theorem 2.12; at the level of the present lemma, the multiplicative contribution is encoded formally by the cusp variables $x_{\mathrm{I}_\bullet}$, $x_{\mathrm{I}^*_\bullet}$, and the correction $P(s; \mathbf{x})$ absorbs the s-structure that is not resolved until the Main Theorem.

Note that for fibers of type I^*_k ($k \geq 1$), the local condition has a dual nature: it imposes *additive* vanishing $\nu_x(a_4) = 2$, $\nu_x(a_6) = 3$ (handled by the sparsity argument of Step 3 below) together with a *cusp* contact order k determined by the discriminant. The additive component enters through J_{add} and is governed by the sparsity threshold $n_1(T)$; the cusp component enters through the cusp variable $x_{\mathrm{I}^*_\bullet}$ and is governed by the motivic discriminant stabilization.

Step 2: Decomposition by additive configuration. An *additive configuration* is a tuple $T = (N_j)_{j \in J_{\mathrm{add}}}$ recording the number of fibers of each additive Kodaira type. We

decompose

$$\mathcal{H}(s;\mathbf{x}) \;=\; \sum_{T} \mathcal{H}_T(s;\mathbf{x}_{\mathrm{mult}}) \cdot \prod_{j\in J_{\mathrm{add}}} x_j^{N_j},$$

where the sum runs over all additive configurations T and $\mathcal{H}_T$ collects, at each height n, the motivic contribution from minimal Weierstrass fibrations with additive configuration T and arbitrary multiplicative fibers (tracked by $\mathbf{x}_{\mathrm{mult}}$).

Step 3: Additive conditions are independent for large n. Fix an additive configuration T. The vanishing conditions $\nu_{x_i}(a_4) \geq p_{j_i}$, $\nu_{x_i}(a_6) \geq q_{j_i}$ at the $|T| := \sum_j N_j$ additive support points impose conditions on the coefficient spaces of total multiplicity $P(T) := \sum_j N_j p_j$ and $Q(T) := \sum_j N_j q_j$ respectively. At distinct points $x_1, \ldots, x_{|T|} \in \mathbb{P}^1$, these are independent linear conditions provided

$$P(T) \;\leq\; 4n+1 \qquad \text{and} \qquad Q(T) \;\leq\; 6n+1, \tag{14}$$

since then each set of vanishing conditions cuts out a codimension-p_{j_i} (resp. q_{j_i}) linear subspace of $H^0(\mathcal{O}(4n))$ (resp. $H^0(\mathcal{O}(6n))$) with disjoint support. Since $P(T)$ and $Q(T)$ depend only on T, while $\dim H^0(\mathcal{O}(4n)) = 4n+1$ and $\dim H^0(\mathcal{O}(6n)) = 6n+1$ grow linearly in n, there exists $n_1(T) \in \mathbb{Z}_{>0}$ such that (14) holds for all $n \geq n_1(T)$.

For $n \geq n_1(T)$, the fiber of $\mathcal{R}_n^{\mu}$ over each point of $\mathrm{Conf}_{|T|}(\mathbb{P}^1)$ decomposes motivically as a product of independent local contributions, each with motivic class A_j as defined above. This yields the Euler product for the *additive* component of $\mathcal{H}_T$ (i.e. the dependence on the additive variables $\{x_\alpha\}_{\alpha\in\mathcal{A}_{\mathrm{nc}}}$ and on s through the additive discriminant degrees c_j) for all $n \geq n_1(T)$. The multiplicative component (dependence on the cusp variables $x_{\mathrm{I}_\bullet}, x_{\mathrm{I}_\bullet^*}$) is formal at this stage, as explained in Step 1.

Step 4: Rationality of each additive slice. Let $R_T(s;\mathbf{x}_{\mathrm{mult}})$ denote the formal power series obtained by expanding the T-component of the right-hand side of (11) (i.e. the Euler product without the correction). By the power-structure formalism (Kapranov's identity applied to each factor), the additive part of R_T is a rational function in s with coefficients in $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}][\mathbf{x}]$. By Step 3, the additive component of $\mathcal{H}_T$ and R_T agree for all s^{12n} with $n \geq n_1(T)$. Therefore

$$\mathcal{H}_T(s;\mathbf{x}_{\mathrm{mult}}) \;=\; R_T(s;\mathbf{x}_{\mathrm{mult}}) \;+\; P_T(s;\mathbf{x}_{\mathrm{mult}})$$

where $P_T \in K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}][\![s,\mathbf{x}_{\mathrm{mult}}]\!]$ is a formal power series that encodes (i) the additive corrections at heights $n < n_1(T)$ and (ii) the full s-structure of the cusp sector, which is not resolved at the $\mathbf{x}$-level. Both contributions are absorbed into $P(s;\mathbf{x})$.

Step 5: Finite assembly. For each fixed monomial $\prod_\alpha x_\alpha^{m_\alpha}$ in $\mathcal{H}(s;\mathbf{x})$, only finitely many additive configurations T contribute (since N_j is bounded by m_α for each α). By Step 4, each $\mathcal{H}_T$ is rational in s and satisfies $\mathcal{H}_T = R_T + P_T$. Summing over all additive configurations, we conclude that

$$\mathcal{H}(s;\mathbf{x}) = \prod_{j\in J}(1-Y_j)^{-\{\mathbb{P}^1\}} + P(s;\mathbf{x})$$

where the coefficient of each monomial in the additive variables $\{x_\alpha\}_{\alpha\in\mathcal{A}_{\mathrm{nc}}}$ is rational in s, and $P(s;\mathbf{x}) = \sum_T P_T(s;\mathbf{x}_{\mathrm{mult}}) \cdot \prod_{j\in J_{\mathrm{add}}} x_j^{N_j}$ is a formal power series in s

and $\mathbf{x}$. The s-structure of P in the cusp variables is not resolved at this stage; see Theorem 2.12.

Finally, for $\alpha \in \{\mathrm{I}_\bullet, \mathrm{I}_\bullet^*\}$ the exponent $\beta_{j,\alpha}$ counts only the number of cusp markings of the given cusp *shape* in factor type j; the individual contact orders k_i are part of the admissible chart data and are handled separately by geometric resummation (Lemma 2.4). ■

Under the specialization $x_\alpha = u^{m(\alpha)-1}$, two types of invisible fibers acquire unbounded multiplicities at fixed u^m: type II (with $m(\mathrm{II}) - 1 = 0$, so $x_{\mathrm{II}} \mapsto 1$) and type I_1 (with $m(\mathrm{I}_1) - 1 = 0$). For type II, the multiplicity is controlled on the non-constant-j locus by $N_{\mathrm{II}} \le \deg(a_4) = 4n$ (Proposition 2.9(1)), while on the isotrivial loci separate exact Euler products apply (Proposition 2.8). For type I_1, the residual simple roots of $\Delta = 4a_4^3 + 27a_6^2$ are handled by the motivic discriminant stabilization (Proposition 2.11).

Proposition 2.8 (Isotrivial Euler products)**.** *Let k be a perfect field with* $\mathrm{char}(k) \neq 2, 3$. *Set $s = t^{1/12}$.*

(1) The $j \equiv 0$ locus. *On $\{a_4 = 0\}$, the Weierstrass model reduces to $y^2 = x^3 + a_6(t)$ with $a_6 \in H^0(\mathbb{P}^1, \mathcal{O}(6n)) \smallsetminus \{0\}$ subject to the minimality constraint $\nu_v(a_6) < 6$ for all v. The discriminant is $\Delta = -432\, a_6^2$, so each singular fiber at v is determined by $\nu_v(a_6) \in \{1, 2, 3, 4, 5\}$:*

$\nu_v(a_6)$	1	2	3	4	5
Fiber type	II	IV	I_0^* (j=0)	IV*	II*
$m_v - 1$	0	2	4	6	8

Since all local conditions are linear *on the single coefficient space $H^0(\mathcal{O}(6n))$, the local-to-global factorization is exact at all heights: no sparsity threshold or surjectivity condition is needed. Define $A_v^{(0)} := (\mathbb{L} - 1)\mathbb{L}^{-\nu}$ for $\nu \in \{1, \dots, 5\}$, and let $w(\nu) := m_\nu - 1$ be the trivial lattice contribution from the table above (explicitly $w(1, \dots, 5) = 0, 2, 4, 6, 8$). Then*

$$Z_{\mathrm{Triv}}^{j=0}(u; t) \;=\; u^2 \cdot \prod_{\nu=1}^{5} \zeta_{\mathbb{P}^1}\big(A_\nu^{(0)}\, u^{w(\nu)}\, s^{2\nu}\big). \tag{15}$$

(2) The $j \equiv 1728$ locus. *On $\{a_6 = 0\}$, the model reduces to $y^2 = x^3 + a_4(t)x$ with $a_4 \in H^0(\mathbb{P}^1, \mathcal{O}(4n)) \smallsetminus \{0\}$ and $\nu_v(a_4) < 4$ for all v. The discriminant is $\Delta = -64\, a_4^3$, and*

$\nu_v(a_4)$	1	2	3
Fiber type	III	I_0^* (j=1728)	III*
$m_v - 1$	1	4	7

All conditions are linear on $H^0(\mathcal{O}(4n))$, and the factorization is again exact:

$$Z_{\mathrm{Triv}}^{j=1728}(u; t) \;=\; u^2 \cdot \prod_{\mu=1}^{3} \zeta_{\mathbb{P}^1}\big(A_\mu^{(1728)}\, u^{w'(\mu)}\, s^{3\mu}\big), \tag{16}$$

with $A_\mu^{(1728)} = (\mathbb{L}-1)\mathbb{L}^{-\mu}$ *and* $w'(1) = 1$, $w'(2) = 4$, $w'(3) = 7$.

Both (15) *and* (16) *are finite products of evaluations of* $\zeta_{\mathbb{P}^1}(y) = 1/\big((1-y)(1-\mathbb{L}\,y)\big)$ *at monomials, hence strictly rational in* s. *Since the discriminant is a pure power of a single Weierstrass coefficient on both loci, every singular fiber is additive and neither the cusp resummation (Lemma 2.4) nor the motivic transversality (Lemma 2.6) is needed.*

Proof. We prove (1); part (2) is identical with $H^0(\mathcal{O}(6n))$ replaced by $H^0(\mathcal{O}(4n))$.

With $a_4 = 0$, the moduli at height n is $H^0(\mathbb{P}^1, \mathcal{O}(6n)) \smallsetminus \{0\}$ modulo $\mathbb{G}_m$, and each local condition $\nu_\nu(a_6) \geq \nu$ is codimension-ν and linear on $H^0(\mathcal{O}(6n))$. For distinct points $\nu_1, \ldots, \nu_r$, these conditions are independent whenever the Hermite interpolation map $H^0(\mathcal{O}(6n)) \to \bigoplus_i J_{\nu_i}^{\nu_i-1}(\mathcal{O}(6n))$ is surjective, which requires $\sum_i \nu_i \leq 6n+1 = \dim H^0(\mathcal{O}(6n))$. Since minimality forces $\nu_i \leq 5$ and $\sum_i 2\nu_i = 12n$ (discriminant degree), one has $\sum_i \nu_i = 6n < 6n+1$, so surjectivity holds at all heights and the Euler product is exact with no correction. The local factor coefficient $A_\nu^{(0)} = (\mathbb{L}-1)\mathbb{L}^{-\nu}$ records the codimension-ν vanishing with nonzero leading term. ■

Proposition 2.9 (Three-locus decomposition)**.** *Decompose the moduli as a disjoint union*

$$\mathcal{W}_n^{\min} = \mathcal{W}_n^{\mathrm{nc}} \sqcup \mathcal{W}_n^{j=0} \sqcup \mathcal{W}_n^{j=1728},$$

where $\mathcal{W}_n^{j=0}$ *parametrizes* $j \equiv 0$ (*equivalently* $a_4 \equiv 0$), $\mathcal{W}_n^{j=1728}$ *parametrizes* $j \equiv 1728$ (*equivalently* $a_6 \equiv 0$), *and* $\mathcal{W}_n^{\mathrm{nc}}$ *is the complement* (*equivalently* $a_4 \not\equiv 0$ *and* $a_6 \not\equiv 0$; *this includes all non-constant* j*-invariant fibrations as well as constant* $j \notin \{0, 1728\}$).

(1) *On the isotrivial loci, the Euler products of Proposition 2.8 are exact at all heights, since all local conditions are linear on a single coefficient space.*
(2) *On* $\mathcal{W}_n^{\mathrm{nc}}$, *a type* II *fiber at* ν *requires* $\nu_\nu(a_4) \geq 1$; *since* $a_4 \not\equiv 0$ *has degree* $\leq 4n$, *the total type-*II *multiplicity satisfies* $N_{\mathrm{II}} \leq 4n$, *so for each fixed* u^m *the additive configuration is bounded and the sparsity argument of Lemma 2.7 gives independence for* $n \geq n_1(T)$.
(3) *At* $u = 1$ *the three loci assemble to the unweighted height zeta function* $Z_{\vec{\lambda}}(t)$, *which equals the exact Euler product of* [BPS22, Thm. 8.9].

Proof. Part (1) is Proposition 2.8. For (2), $\sum_\nu \nu_\nu(a_4) \leq \deg(a_4) = 4n$ gives $N_{\mathrm{II}} \leq 4n$, so all additive multiplicities contributing to a fixed u^m are bounded and $P(T) \leq 4n+1$ holds for $n \geq n_1(T)$. Part (3) follows from [BPS22, Thm. 8.9]. ■

Motivic discriminant stabilization. The motivic transversality (Lemma 2.6) establishes multi-point independence of discriminant conditions when the jet evaluation map is surjective: $\sum_i (k_i+1) \leq 4n+1$. For a purely multiplicative fibration at height n, the discriminant budget gives $\sum_i k_i = 12n$, so the total jet demand is $\sum_i (k_i+1) = 12n + r \geq 12n+1$, which exceeds the surjectivity bound $4n+1$ for every $n \geq 1$. More generally, for a fixed additive configuration together with cusp fibers consuming a fraction $> \frac{1}{3}$ of the discriminant degree, the surjectivity condition fails. Since the generic elliptic surface over $\mathbb{P}^1$ is semistable (all singular fibers multiplicative), the motivic transversality alone does not cover the dominant part

of moduli. To handle this regime, we adapt the motivic stabilization method of Vakil–Wood [VW15] to the discriminant $\Delta = 4a_4^3 + 27a_6^2$ of the Weierstrass model. The key idea is a truncated inclusion–exclusion in the dimensional completion of the Grothendieck ring, where the error from discriminant root multiplicities becomes dimensionally negligible.

The dimensional completion. Recall [VW15, Section 1.4] that the *dimensional filtration* on $\mathcal{M}_{\mathbb{L}} := K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}]$ is defined by

$$F^d\mathcal{M}_{\mathbb{L}} := \left\langle [X]\mathbb{L}^{-n} : \dim X - n \le -d \right\rangle.$$

Let $\widehat{\mathcal{M}_{\mathbb{L}}}$ denote the completion of $\mathcal{M}_{\mathbb{L}}$ with respect to this filtration. This is Kontsevich's ring of motivic integrals [Kon95]; it inherits a ring structure [VW15, §1.4]. A sequence $a_N \to 0$ in $\widehat{\mathcal{M}_{\mathbb{L}}}$ means $\dim(a_N) \to -\infty$.

Visible and invisible fibers. For the u-extraction, we call a singular fiber *visible* if $m(F_v) - 1 \ge 1$, i.e. if it contributes positively to the trivial lattice beyond the minimal value 2. On the non-constant-j locus $\mathcal{W}_n^{\mathrm{nc}}$, the visible fiber types are:

- *additive visible:* III, IV, I_0^*, IV*, III*, II* (each with $m_v - 1 \ge 1$);
- *multiplicative visible:* I_k with $k \ge 2$ (with $m_v - 1 = k - 1 \ge 1$);
- *starred cusp:* I_k^* with $k \ge 1$ (with $m_v - 1 = k + 4 \ge 5$).

The *invisible* types are II (with $m_v - 1 = 0$) and I_1 (with $m_v - 1 = 0$).

For a fixed power u^m in Z_{Triv}, the visible fibers have total weight $\sum(m_v - 1) = m - 2$. Since each visible fiber contributes weight ≥ 1, the number of visible fibers is at most $m - 2$. Their types and (unordered) positions on $\mathbb{P}^1$ form a *visible configuration* σ, with weight $|\sigma|_{\mathrm{wt}} := \sum_{v\in\sigma}(m_v - 1) = m - 2$; there are finitely many such σ for each m.

Bounded visible transversality. Fix a visible configuration σ with $|\sigma|_{\mathrm{wt}} = m - 2$. Each visible fiber imposes jet conditions at its support point: an additive visible fiber of type Θ requires $v_v(a_4) \ge p_\Theta$ and $v_v(a_6) \ge q_\Theta$ conditions on the Weierstrass coefficients, while a multiplicative visible fiber I_{k_i} (with $k_i \ge 2$) or the cusp component of $\mathrm{I}_{k_j}^*$ requires prescribing $(k_i + 1)$ or $(k_j + 1)$ jets of (a_4, a_6) respectively. Write

$$D_{a_4}(\sigma) := \sum_{v\in\sigma_{\mathrm{add}}} p_{\Theta_v} + \sum_{v\in\sigma_{\mathrm{mult}}} (k_v + 1), \qquad D_{a_6}(\sigma) := \sum_{v\in\sigma_{\mathrm{add}}} q_{\Theta_v} + \sum_{v\in\sigma_{\mathrm{mult}}} (k_v + 1)$$

for the total jet demands on $H^0(\mathcal{O}(4n))$ and $H^0(\mathcal{O}(6n))$ respectively, and $D(\sigma) := D_{a_4}(\sigma) + D_{a_6}(\sigma)$. Then

$$D(\sigma) \le D(m) := \max_{\sigma'\in\Sigma(m)} D(\sigma') < \infty, \tag{17}$$

since $\Sigma(m)$ is finite: each visible fiber contributes weight ≥ 1 to the total $m - 2$, bounding the number of visible fibers, while the weight constraint $\sum(k_i - 1) \le m - 2$ bounds the total multiplicative jet demand, and the additive jet demands are bounded by the fiber types in σ.

By Lemma 2.6 (for the multiplicative visible fibers) and the sparsity argument (Lemma 2.7, for the additive visible fibers), the jet evaluation map at the σ-support points is surjective for $n \ge n_0(m) := \lceil (D(m) - 1)/4 \rceil$.

Let

$$U_n^{\mathrm{nc}} \coloneqq \big\{(a_4, a_6) \in H^0\big(\mathcal{O}(4n)\big) \oplus H^0\big(\mathcal{O}(6n)\big) : \, a_4 \not\equiv 0, \; a_6 \not\equiv 0\big\}$$

denote the non-isotrivial coprime locus, and define

$$G_\sigma(n) \coloneqq \big\{(a_4, a_6) \in U_n^{\mathrm{nc}} : \text{visible conditions of } \sigma \text{ hold at unordered support points}\big\}.$$

For $n \geq n_0(m)$, the motivic class of $G_\sigma(n)$ factors as

$$\big\{G_\sigma(n)\big\} \;=\; C(\sigma) \cdot \mathbb{L}^{10n+2-D(\sigma)}, \tag{18}$$

where $C(\sigma) \in \mathcal{M}_{\mathbb{L}}$ depends on σ alone (it incorporates the local fiber classes and the configuration-space class).

The overcounting problem. The locus $G_\sigma(n)$ includes Weierstrass data whose *exact* visible configuration is a strict enlargement $\sigma' \supsetneq \sigma$: the data may have extra visible fibers at points outside the σ-support. Such data have $T(S) > m$ and contribute to $u^{m'}$ with $m' > m$, not to $[u^m]Z_{\mathrm{Triv}}$. To extract the exact visible locus

$$F_\sigma(n) \coloneqq \big\{(a_4, a_6) \in G_\sigma(n) : \text{no visible fibers outside the } \sigma\text{-support}\big\},$$

we must remove the extra-visible-fiber contributions.

On the coprime locus outside the σ-support, the condition "no visible multiplicative fiber at v" is $\mathrm{ord}_v(\Delta) \leq 1$ (i.e., Δ has at most a simple root at v). Equivalently, the residual discriminant must have *no multiple roots* on $\mathbb{P}^1 \smallsetminus V_\sigma$, where V_σ is the σ-support. This is a motivic discriminant-squarefreeness condition on the image of the nonlinear map $\Delta = 4a_4^3 + 27a_6^2$, which we handle by the Vakil–Wood method.

(Extra visible *additive* fibers cannot occur on the coprime locus outside V_σ: every additive type has $\nu_v(a_4) \geq 1$ and $\nu_v(a_6) \geq 1$, forcing $(a_4(v), a_6(v)) = (0,0)$.)

Local codimension of a multiple root. We compute the codimension of the "extra multiple root" condition at a single coprime point v.

Lemma 2.10. *Let $v \in \mathbb{P}^1 \smallsetminus V_\sigma$. On the coprime locus $\{(a_4(v), a_6(v)) \neq (0,0)\}$, the condition $\mathrm{ord}_v(\Delta) \geq 2$ defines a constructible subset of the 1-jet space $J_v^1(\mathcal{O}(4n)) \oplus J_v^1(\mathcal{O}(6n)) \cong \mathbb{A}^4$ of class $\mathbb{L}^2 - \mathbb{L}$ and codimension 2.*

Proof. Write $\alpha = a_4(v)$, $\alpha' = a_4'(v)$, $\beta = a_6(v)$, $\beta' = a_6'(v)$ for the 1-jet coordinates. The condition $\mathrm{ord}_v(\Delta) \geq 2$ on the coprime locus $(\alpha, \beta) \neq (0,0)$ is:

(i) $\Delta(v) = 4\alpha^3 + 27\beta^2 = 0$: this restricts (α, β) to the cuspidal cubic $\{4x^3 + 27y^2 = 0\} \smallsetminus \{(0,0)\}$, which is parametrized by $(\alpha, \beta) = (-3\lambda^2, 2\lambda^3)$ for $\lambda \in \mathbb{G}_m$. Its class is $\mathbb{L} - 1$.

(ii) $\Delta'(v) = 12\alpha^2\alpha' + 54\beta\beta' = 0$: given (i), we have $\beta \neq 0$ (since $4\alpha^3 + 27\beta^2 = 0$ with $(\alpha, \beta) \neq (0,0)$ forces $\beta \neq 0$). Hence $12\alpha^2\alpha' + 54\beta\beta' = 0$ is a nontrivial *linear* condition on (α', β'), cutting codimension 1 in $\mathbb{A}^2$. Its solution space has class $\mathbb{L}$.

The total class in $\mathbb{A}^4 \supset \{(\alpha, \alpha', \beta, \beta')\}$ is $(\mathbb{L}-1)\cdot\mathbb{L} = \mathbb{L}^2 - \mathbb{L}$, which has dimension 2 inside the ambient $\mathbb{A}^4$ (dimension 4). The codimension is 2. ∎

Truncated inclusion–exclusion. Following [VW15, Section 3], we write

$$F_\sigma(n) \;=\; G_\sigma(n) - G_{\sigma,\geq 1}(n) \;=\; G_\sigma(n) - \sum_{w \in \mathbb{P}^1 \smallsetminus V_\sigma} G^{\geq}_{\sigma \cup \{w\}}(n) + G_{\sigma,\geq 2}(n) \;=\; \cdots, \tag{19}$$

where $G_{\sigma,\geq k}(n)$ denotes the sublocus of $G_\sigma(n)$ where the residual discriminant has at least k additional multiple roots (each of multiplicity ≥ 2) on $\mathbb{P}^1 \smallsetminus V_\sigma$.

Since $G_{\sigma,\geq k}$ is the image of $G^{\geq}_{\sigma\cup\tau}$ (with $|\tau| = k$) under the projection forgetting the τ-labels, we have the Vakil–Wood decomposition

$$F_\sigma(n) \;=\; \sum_{\ell=0}^{N}(-1)^\ell \sum_{|\tau|=\ell}\big\{G^{\geq}_{\sigma\cup\tau}(n)\big\} \;+\; (-1)^{N+1}\big\{G_{\sigma,\geq N+1}(n)\big\}.$$

For each $\ell \leq N$, the locus $G^{\geq}_{\sigma\cup\tau}(n)$ parametrizes coprime data satisfying the σ-conditions and having $\mathrm{ord}_{w_i}(\Delta) \geq 2$ at ℓ additional unordered points $w_1,\ldots,w_\ell$. The 1-jet evaluation map at the σ-support and the τ-points,

$$\mathrm{ev}\colon H^0(\mathcal{O}(4n)) \oplus H^0(\mathcal{O}(6n)) \;\longrightarrow\; \bigoplus_{v\in V_\sigma}\big(J_v^{k_v} \oplus J_v^{k_v}\big) \;\oplus\; \bigoplus_{i=1}^{\ell}\big(J^1_{w_i} \oplus J^1_{w_i}\big),$$

has source dimension $10n+2$ and target dimension $D(\sigma)+4\ell$. This is surjective whenever

$$D_{a_4}(\sigma)+2\ell \;\leq\; 4n+1 \qquad \text{and} \qquad D_{a_6}(\sigma)+2\ell \;\leq\; 6n+1. \tag{20}$$

When surjective, Lemma 2.10 gives the factorization

$$\big\{G^{\geq}_{\sigma\cup\tau}(n)\big\} \;=\; C(\sigma)\cdot\big\{\mathrm{Sym}^\ell(\mathbb{P}^1 \smallsetminus V_\sigma)\big\}\cdot(\mathbb{L}^2-\mathbb{L})^\ell\cdot\mathbb{L}^{10n+2-D(\sigma)-4\ell}. \tag{21}$$

The first factor $C(\sigma)$ is the local class at the σ-points (same as in (18)); the second factor chooses the ℓ extra positions; the third is the local multiple-root class from Lemma 2.10 at each extra point; the fourth is the kernel dimension.

Proposition 2.11 (Motivic discriminant stabilization)**.** *For each visible configuration σ with $|\sigma|_{\mathrm{wt}} = m-2$, and each truncation depth $N \geq 0$, the truncated inclusion–exclusion*

$$F^{(N)}_\sigma(n) \;:=\; \sum_{\ell=0}^{N}(-1)^\ell \sum_{|\tau|=\ell}\big\{G^{\geq}_{\sigma\cup\tau}(n)\big\} \tag{22}$$

satisfies the following for all

$$n \;\geq\; n_0(\sigma,N) \;:=\; \max\Big(\Big\lceil\frac{D_{a_4}(\sigma)+2N}{4}\Big\rceil, \Big\lceil\frac{D_{a_6}(\sigma)+2N}{6}\Big\rceil\Big),$$

where $D_{a_4}(\sigma)+D_{a_6}(\sigma) = D(\sigma) \leq D(m)$:

(1) *Each summand in* (22) *is given by* (21)*. The three factors contribute dimensions ℓ, 2ℓ, and $10n+2-D(\sigma)-4\ell$ respectively, for a total of $10n+2-D(\sigma)-\ell$.*

(2) *The truncation error satisfies the dimension bound*

$$\dim\big(F_\sigma(n)-F^{(N)}_\sigma(n)\big) \;\leq\; 10n+2-D(\sigma)-(N+1). \tag{23}$$

(3) *After normalizing by $\mathbb{L}^{10n+2-D(\sigma)}$, the error lies in $F^{N+1}\widehat{\mathcal{M}}_{\mathbb{L}}$. In particular,*

$$\frac{\{F_\sigma(n)\}}{\mathbb{L}^{10n+2-D(\sigma)}} \;\longrightarrow\; C(\sigma)\cdot\zeta_{\mathbb{P}^1\smallsetminus V_\sigma}\big(-\mathbb{L}^{-3}(\mathbb{L}-1)\big)$$

in $\widehat{\mathcal{M}}_{\mathbb{L}}$ as $N\to\infty$, for each $n \geq n_0(\sigma,N)$, where $\zeta_X(y) = \sum_{N\geq 0}\{\mathrm{Sym}^N(X)\}\,y^N$ is the Kapranov motivic zeta function.

Proof. Part (1) follows from the surjectivity condition (20): for $\ell \le N$ and $n \ge n_0(\sigma, N)$, the jet evaluation is surjective and (21) applies.

For part (2), the error is $(-1)^{N+1}\{G_{\sigma,\ge N+1}(n)\}$, where $G_{\sigma,\ge N+1}(n)$ parametrizes coprime data satisfying the σ-conditions and having at least $N+1$ additional multiple roots of Δ on $\mathbb{P}^1 \smallsetminus V_\sigma$. Consider the incidence variety

$$\mathcal{I}_{N+1} := \big\{\big((a_4, a_6), (w_1, \dots, w_{N+1})\big) \in G_\sigma(n) \times (\mathbb{P}^1 \smallsetminus V_\sigma)^{N+1} : \operatorname{ord}_{w_i}(\Delta) \ge 2 \ \ \forall\, i\big\}.$$

Since $G_{\sigma,\ge N+1}(n)$ is the image of $\mathcal{I}_{N+1}$ under the first projection (we use the ordered product $(\mathbb{P}^1 \smallsetminus V_\sigma)^{N+1}$; passing to Sym^{N+1} does not change the dimension), $\dim G_{\sigma,\ge N+1} \le \dim \mathcal{I}_{N+1}$. We estimate $\dim \mathcal{I}_{N+1}$ via the second projection $\pi_2 \colon \mathcal{I}_{N+1} \to (\mathbb{P}^1 \smallsetminus V_\sigma)^{N+1}$.

Fix a *generic* configuration $w = (w_1, \dots, w_{N+1})$ with w_i pairwise distinct and $w_i \notin V_\sigma$. The fiber $\pi_2^{-1}(w)$ consists of $(a_4, a_6) \in G_\sigma(n)$ satisfying $\Delta(w_i) = 0$ and $\Delta'(w_i) = 0$ at each w_i. At w_i, the condition $\Delta(w_i) = 4a_4(w_i)^3 + 27a_6(w_i)^2 = 0$ is a single polynomial equation in the linear forms $a_4(w_i)$, $a_6(w_i)$ on the source. Since $4X^3+27Y^2$ is irreducible and nonconstant on $\{(X,Y) \ne (0,0)\}$, this condition cuts codimension 1 on the coprime locus. Moreover, for generic w_i, the evaluation functionals $a_4(w_i)$, $a_6(w_i)$ are independent of the evaluation functionals at w_j ($j \ne i$) and of the σ-jet constraints, since distinct-point evaluations on $H^0(\mathcal{O}(d))$ are linearly independent functionals for any $d \ge 0$. Hence $\Delta(w_i) = 0$ imposes a fresh codimension-1 condition at each w_i. Given $\Delta(w_i) = 0$, the derivative condition $\Delta'(w_i) = 12a_4(w_i)^2\, a_4'(w_i) + 54a_6(w_i)\, a_6'(w_i) = 0$ is a nontrivial linear equation in $a_4'(w_i)$, $a_6'(w_i)$ (by Lemma 2.10, since $(a_4(w_i), a_6(w_i)) \ne (0,0)$ on the coprime locus); for generic w_i this derivative evaluation is independent of the previous constraints by the same distinct-point interpolation argument, cutting a further codimension 1. Therefore the $2(N+1)$ conditions at $w_1, \dots, w_{N+1}$ are generically independent, giving

$$\dim \pi_2^{-1}(w) \ \le\ (10n + 2 - D(\sigma)) - 2(N+1).$$

By upper semicontinuity of fiber dimension ([Har77, Ex. II.3.22]), the generic fiber dimension bounds the dimension of the total space:

$$\dim \mathcal{I}_{N+1} \ \le\ (10n + 2 - D(\sigma) - 2(N+1)) + (N+1) \ =\ 10n + 2 - D(\sigma) - (N+1),$$

and hence $\dim G_{\sigma,\ge N+1} \le \dim \mathcal{I}_{N+1} \le 10n + 2 - D(\sigma) - (N+1)$, yielding (23). The bound holds vacuously if $G_{\sigma,\ge N+1}(n)$ is empty.

Part (3) follows: after normalizing by $\mathbb{L}^{10n+2-D(\sigma)}$, the error has dimension $\le -(N+1)$, hence lies in $F^{N+1}\widehat{\mathcal{M}}_{\mathbb{L}}$ and converges to 0 as $N \to \infty$. The limit is the full inclusion–exclusion sum, which converges in $\widehat{\mathcal{M}}_{\mathbb{L}}$ to the stated expression.

This adapts [VW15, Section 3] to the nonlinear discriminant $\Delta = 4a_4^3 + 27a_6^2$: the role of the growing line bundle in [VW15] is played by the growing height n, and the limit $\zeta_{\mathbb{P}^1 \smallsetminus V_\sigma}(-\alpha)$ with $\alpha = \mathbb{L}^{-3}(\mathbb{L}-1)$ is the motivic analog of the squarefree probability of [VW15, Theorem 1.13]. ∎

The main theorem.

Theorem 2.12 (Approximate rationality of Z_{Triv} over $k(t)$)**.** *Let k be a perfect field with* $\mathrm{char}(k) \neq 2, 3$. *Set $K = k(t)$ and $s = t^{1/12}$ (so $t = s^{12}$). Then for each $m \geq 2$,*

$$[u^m]Z_{\mathrm{Triv}}(u;t) \in \overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)} \subset \widehat{\mathcal{M}}_{\mathbb{L}}[\![s]\!],$$

where $\widehat{\mathcal{M}}_{\mathbb{L}}$ is the completion of $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}]$ with respect to the dimensional filtration, and $\overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}$ denotes the closure of the rational functions $\widehat{\mathcal{M}}_{\mathbb{L}}(s)$ inside $\widehat{\mathcal{M}}_{\mathbb{L}}[\![s]\!]$ with respect to the coefficient-wise dimensional filtration topology.

More precisely, for each $m \geq 2$ and each truncation depth $N \geq 0$, there exists a rational function $R_{m,N}(s) \in \widehat{\mathcal{M}}_{\mathbb{L}}(s)$ such that the coefficient of s^{12n} in $[u^m]Z_{\mathrm{Triv}}(u;t) - R_{m,N}(s)$ has dimension $\leq 10n - (N+1)$ for all $n \geq n_0(m,N)$, where $n_0(m,N) := \max_{\sigma \in \Sigma(m)}\big(n_0(\sigma,N), n_1(\sigma)\big)$ is the surjectivity threshold of Proposition 2.11 taken over the finitely many visible configurations σ of weight $m-2$.

The approximating rational functions are constructed via the three-locus decomposition of Proposition 2.9 together with the motivic discriminant stabilization of Proposition 2.11:

(i) ***Isotrivial loci.*** *On $j \equiv 0$ and $j \equiv 1728$, the discriminant reduces to a pure power of a single Weierstrass coefficient, all local conditions are linear, and the Hermite interpolation map is surjective at all heights, yielding exact Euler products with no correction and no approximation needed (Proposition 2.8).*

(ii) ***Non-constant-j locus, additive sector.*** *On $\mathcal{W}_n^{\mathrm{nc}}$, type-*II *multiplicity is bounded by $\deg(a_4) = 4n$ (Proposition 2.9(1)), and the remaining additive conditions are linear with bounded total codimension at each fixed u^m. The sparsity argument gives an exact Euler product for the additive factors at all heights above an explicit threshold.*

(iii) ***Non-constant-j locus, multiplicative residual.*** *The residual multiplicative sector (discriminant root multiplicities of $\Delta = 4a_4^3 + 27a_6^2$ at coprime points) is handled by the motivic discriminant stabilization (Proposition 2.11): for each visible configuration σ, the truncated inclusion–exclusion at depth N produces a rational approximation with error in $F^{N+1}\widehat{\mathcal{M}}_{\mathbb{L}}$.*

At $u = 1$ the trivial lattice grading is forgotten, all three loci assemble to the motivic height zeta function $Z_{\vec{\lambda}}(t)$ of [BPS22]*, and the approximation becomes exact: $Z_{\mathrm{Triv}}(1;t) \in K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}](s)$.*

Proof. Work in the dimensional completion $\widehat{\mathcal{M}}_{\mathbb{L}}$ of $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}]$.

Height-zero term. For $n = 0$ the discriminant degree is 0, hence the corresponding elliptic curve over $K = k(t)$ has everywhere good reduction and is therefore constant. Equivalently, $\mathcal{W}_0^{\min}$ identifies with the moduli stack of smooth elliptic curves,

$$\mathcal{W}_0^{\min} \cong \mathcal{M}_{1,1}.$$

Therefore $\{\mathcal{W}_0^{\min}\} = \mathbb{L}$ as in [Eke25], and the required constant term is $u^2 \cdot \mathbb{L}$.

Three-locus decomposition. As in Proposition 2.9, decompose $Z_{\mathrm{Triv}} = Z_{\mathrm{Triv}}^{\mathrm{nc}} + Z_{\mathrm{Triv}}^{j=0} + Z_{\mathrm{Triv}}^{j=1728}$.

Isotrivial loci. By Proposition 2.8, the isotrivial Euler products for $Z_{\mathrm{Triv}}^{j=0}$ and $Z_{\mathrm{Triv}}^{j=1728}$ are exact at all heights. These are finite products of Kapranov factors, hence rational in s already in $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}](s)$.

Non-constant-j locus: visible/invisible decomposition. Fix $m \geq 2$ and extract $[u^m]Z^{\mathrm{nc}}_{\mathrm{Triv}}$. On $\mathcal{W}^{\mathrm{nc}}_n$, both $a_4 \not\equiv 0$ and $a_6 \not\equiv 0$. Each Weierstrass datum $(a_4, a_6) \in U^{\mathrm{nc}}_n$ determines a fiber configuration with trivial lattice rank $T(S) = 2 + \sum_v (m_v - 1)$. The *visible* fibers (those with $m_v - 1 \geq 1$: additive types $\mathrm{III}, \mathrm{IV}, \mathrm{I}^*_0, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$; multiplicative types I_k with $k \geq 2$; and starred cusp types I^*_k with $k \geq 1$) have total weight $\sum(m_v - 1) = m - 2$. The *invisible* fibers (II and I_1, each with $m_v - 1 = 0$) contribute nothing to u^m.

Let $\Sigma(m)$ be the finite set of visible configurations σ with $|\sigma|_{\mathrm{wt}} = m - 2$. Then

$$[u^m]Z^{\mathrm{nc}}_{\mathrm{Triv}} = \sum_{\sigma \in \Sigma(m)} \sum_{n \geq 1} \frac{\{F_\sigma(n)\}}{\{\mathrm{PGL}_2\}} s^{12n}, \tag{24}$$

where $F_\sigma(n)$ is the locus of coprime data in U^{nc}_n with visible configuration *exactly* σ (at unordered support points on $\mathbb{P}^1$), and PGL_2 accounts for the Weierstrass automorphism. The sum over σ is finite (depending on m). Here the PGL_2 quotient corresponds to the unparameterized base $\mathbb{P}^1$; the isotrivial Euler products of Proposition 2.8 use the parameterized convention (modulo $\mathbb{G}_m$), the two being related by $\{\mathrm{PGL}_2\} = \mathbb{L}(\mathbb{L}^2 - 1) = \{\mathbb{G}_m\} \cdot \mathbb{L}(\mathbb{L} + 1)$.

Additive visible fibers: bounded transversality. For each $\sigma \in \Sigma(m)$, the additive part of σ (types $\mathrm{III}, \mathrm{IV}, \mathrm{I}^*_0, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ and the additive component of I^*_k) imposes linear vanishing conditions $\nu_v(a_4) \geq p$, $\nu_v(a_6) \geq q$ at the support points. On the non-constant-j locus, type-II fibers satisfy $\nu_v(a_4) \geq 1$; since $a_4 \not\equiv 0$ has degree $\leq 4n$, the total type-II multiplicity is at most $4n$. Since II is invisible, the additive visible fibers have bounded total vanishing order $P(\sigma) + Q(\sigma) \leq D(m)$ (cf. (17)). By the sparsity argument (Lemma 2.7, Step 3), these conditions are independent for $n \geq n_1(\sigma)$.

Multiplicative visible fibers: bounded transversality. The multiplicative visible part of σ consists of fibers I_{k_i} with $k_i \geq 2$ and the cusp component of $\mathrm{I}^*_{k_j}$. Since $\sum(k_i - 1) + \sum(k_j + 4) \leq m - 2$, the total jet demand from visible multiplicative fibers satisfies $\sum(k_i + 1) + \sum(k_j + 1) \leq D(m)$, a bound depending on m alone. By the motivic transversality (Lemma 2.6), the jet evaluation map at the visible support points is surjective for $n \geq n_0(\sigma) := \max\big(\lceil D_{a_4}(\sigma)/4 \rceil, \lceil D_{a_6}(\sigma)/6 \rceil\big)$.

For $n \geq \max(n_0(\sigma), n_1(\sigma))$, the class $\{G_\sigma(n)\}$ factors as in (18):

$$\{G_\sigma(n)\} = C(\sigma) \cdot \mathbb{L}^{10n+2-D(\sigma)},$$

where $G_\sigma(n) \supseteq F_\sigma(n)$ includes data with possible extra visible fibers, and $C(\sigma)$ depends on σ alone.

Multiplicative residual: Vakil–Wood stabilization. To pass from $G_\sigma(n)$ to $F_\sigma(n)$, we remove the extra visible fibers at points outside the σ-support. By Proposition 2.11, for each truncation depth N and $n \geq n_0(m, N)$, the truncated approximation $F^{(N)}_\sigma(n)$ satisfies (21) and the error $F_\sigma(n) - F^{(N)}_\sigma(n)$ has dimension $\leq 10n + 2 - D(\sigma) - (N+1)$. The generating function of the truncated approximation, $\sum_{n \geq n_0} F^{(N)}_\sigma(n) s^{12n}$, is rational in s (each summand in (22) is a product of a motivic constant, a power of $\mathbb{L}^n$, and a polynomial in $\mathbb{L}^n$ from the Sym^ℓ factor, hence contributes a rational function of $\mathbb{L}^{10} s^{12}$). The error series $\sum_n (F_\sigma(n) - F^{(N)}_\sigma(n)) s^{12n}$ has

coefficients of dimension $\leq 10n+2-D(\sigma)-(N+1)$. Therefore, for each N, the generating function $\sum_n \{F_\sigma(n)\}s^{12n}$ is within dimensional distance $N+1$ of a rational function of s in $\widehat{\mathcal{M}}_{\mathbb{L}}[\![s]\!]$. Since N is arbitrary, this shows $\sum_n \{F_\sigma(n)\}s^{12n} \in \overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}$.

Small-height terms. For $n < \max(n_0(m,N), n_1(\sigma))$ (finitely many heights, depending on m), the classes $\{F_\sigma(n)\}$ are arbitrary elements of $K_0(\mathrm{Stck}_k)$. Their contributions $\sum_{n<n_0}\{F_\sigma(n)\}s^{12n}$ form a polynomial in s, which is rational.

Assembly. For each $\sigma \in \Sigma(m)$, the generating function $\sum_n \{F_\sigma(n)\}s^{12n}$ lies in $\overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}$ (large heights by the stabilization, small heights by polynomiality). Summing over the finite set $\Sigma(m)$ and dividing by $\{\mathrm{PGL}_2\}$, we conclude that $[u^m]Z^{\mathrm{nc}}_{\mathrm{Triv}} \in \overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}$.

Combined with the isotrivial Euler products (rational in s over $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}]$, hence in particular in $\overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}$) and the height-zero term, we obtain $[u^m]Z_{\mathrm{Triv}} \in \overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}$ for each m.

Vanishing at $u=1$. At $u=1$, the trivial lattice grading is forgotten: $Z_{\mathrm{Triv}}(1;t) = Z_{\vec{\lambda}}(t)$, the unweighted motivic height zeta function. By [BPS22, Thm. 8.9], this is computed as a single exact Euler product. In particular, the stabilization of Proposition 2.11 is exact at $u=1$: every visible configuration contributes with weight 1, so the overcounting from G_σ cancels in the sum over all σ, and the correction vanishes. ■

As a consequence of Theorem 2.12 and the weight bound in mixed Hodge theory, over $k=\mathbb{C}$ the approximate motivic rationality yields strict rationality of the Hodge–Deligne realization.

Theorem 2.13 (Rationality of the Hodge–Deligne realization over $\mathbb{C}$)**.** *Let $k=\mathbb{C}$, $K=\mathbb{C}(t)$, and $s=t^{1/12}$. Denote by*

$$\mathrm{HD}\colon K_0(\mathrm{Stck}_{\mathbb{C}})[\mathbb{L}^{-1}] \longrightarrow \mathbb{Z}[x,y,(xy)^{-1}], \qquad \mathbb{L}\mapsto xy,$$

the Hodge–Deligne motivic measure [Eke25]*, which extends the compactly supported virtual Hodge numbers to algebraic stacks. Then for each $m \geq 2$, the generating function*

$$\sum_{n\geq 0} \mathrm{HD}\big(\{\mathcal{W}_n^{\min}(m)\}\cdot \mathbb{L}^{-10n}\big)s^{12n} \in \mathbb{Z}[x,y,(xy)^{-1}](s)$$

is a rational function of s, with no completion required. More precisely, there exists a Laurent polynomial $K_m \in \mathbb{Z}[x,y,(xy)^{-1}]$ such that for each $(p_0,q_0)\in\mathbb{Z}^2$, setting $N = \lceil -(p_0+q_0)/2 \rceil$, one has for all $n \geq n_0(m,N)$ (the surjectivity threshold of Proposition 2.11),

$$e_c^{p_0,q_0}\big(\{\mathcal{W}_n^{\min}(m)\}\cdot \mathbb{L}^{-10n}\big) = [x^{p_0}y^{q_0}]K_m. \tag{25}$$

Proof. The key property of the Hodge–Deligne measure is the *weight bound*: for $\alpha \in K_0(\mathrm{Stck}_{\mathbb{C}})[\mathbb{L}^{-1}]$ with $\dim(\alpha)\leq d$, one has $e^{p,q}(\alpha)=0$ for $p+q>2d$. This holds for elements of $K_0(\mathrm{Stck}_{\mathbb{C}})[\mathbb{L}^{-1}]$ by reduction to the variety case: every such element is a finite $\mathbb{Z}[\mathbb{L}^{-1}]$-linear combination of variety classes via the scissor relations (cf. [Eke25, Thm. 1.2]), the Hodge–Deligne measure respects these relations, and Deligne's weight bound ([Del74]) applies termwise.

Step 1: Structure of the VW approximation. Fix $m \geq 2$ and a visible configuration $\sigma \in \Sigma(m)$. By Proposition 2.11, for each truncation depth $N \geq 0$ and $n \geq n_0(m,N)$,

$$\frac{\{F_\sigma^{(N)}(n)\}}{\mathbb{L}^{10n+2-D(\sigma)}} = C(\sigma)\sum_{\ell=0}^{N}(-1)^\ell \left\{\mathrm{Sym}^\ell(\mathbb{P}^1 \smallsetminus V_\sigma)\right\} \cdot \left(\mathbb{L}^{-3}(\mathbb{L}-1)\right)^\ell =: K_{\sigma,N},$$

where $K_{\sigma,N}$ is *independent of* n: the n-dependence in (21) enters only through the kernel factor $\mathbb{L}^{10n+2-D(\sigma)-4\ell}$, which is absorbed by the normalization.

Step 2: Exact vanishing of the error under HD**.** By Proposition 2.11(2), the normalized error $E_{\sigma,N}(n) := \left(F_\sigma(n) - F_\sigma^{(N)}(n)\right) \cdot \mathbb{L}^{-(10n+2-D(\sigma))}$ satisfies $\dim(E_{\sigma,N}(n)) \leq -(N+1)$. By the weight bound, $e^{p,q}(E_{\sigma,N}(n)) = 0$ for all $p+q > -2(N+1)$.

Step 3: Coefficient-wise stabilization. Fix $(p_0,q_0) \in \mathbb{Z}^2$ and choose $N \geq \lceil -(p_0+q_0)/2 \rceil$, so that $p_0+q_0 > -2(N+1)$. Then:

(i) By Step 2, the error $E_{\sigma,N}(n)$ does not contribute to bidegree (p_0,q_0) under HD: the Hodge number is identically zero, not merely small.

(ii) By Step 1, the approximation $K_{\sigma,N}$ at bidegree (p_0,q_0) is a fixed integer, independent of n (for $n \geq n_0(m,N)$).

(iii) The Hodge–Deligne image of the limit $\zeta_{\mathbb{P}^1 \smallsetminus V_\sigma}(-\alpha)$ with $\alpha = \mathbb{L}^{-3}(\mathbb{L}-1)$ is a *finite* Laurent polynomial in $\mathbb{Z}[x,y,(xy)^{-1}]$. To see this, note that $\mathrm{HD}(\alpha) = (xy)^{-2} - (xy)^{-3}$ has maximum total weight -4, and $\mathrm{HD}(\mathrm{Sym}^\ell(\mathbb{P}^1 \smallsetminus V_\sigma))$ has maximum total weight 2ℓ, so $\mathrm{HD}(\mathrm{Sym}^\ell) \cdot \mathrm{HD}(\alpha)^\ell$ has Hodge numbers supported in total weight $\leq 2\ell - 4\ell = -2\ell$. At any fixed bidegree (p_0,q_0), only finitely many ℓ with $-2\ell \geq p_0+q_0$ contribute. Therefore $[x^{p_0}y^{q_0}]K_{\sigma,N}$ stabilizes for N sufficiently large.

Combining (i)–(iii): for each (p_0,q_0), the Hodge number $e^{p_0,q_0}(F_\sigma(n) \cdot \mathbb{L}^{-(10n+2-D(\sigma))})$ equals $[x^{p_0}y^{q_0}]K_{\sigma,N}$ for all $n \geq n_0(m,N)$, and this value is independent of N.

The full coefficient $\left\{\mathcal{W}_n^{\min}(m)\right\} \cdot \mathbb{L}^{-10n}$ is assembled from the finite sum over $\sigma \in \Sigma(m)$ (each handled above), the isotrivial Euler products (which are strictly rational in s by Proposition 2.8, hence contribute eventually constant Hodge numbers), and finitely many small-height corrections (a polynomial in s). Summing, we obtain (25) with K_m the Hodge–Deligne polynomial of the assembled limiting class.

Step 4: Rationality in s**.** By (25), for each (p_0,q_0) the sequence $n \mapsto e_c^{p_0,q_0}\left([u^m]\, Z_{\mathrm{Triv}}\big|_{t^n} \cdot \mathbb{L}^{-10n}\right)$ is eventually constant. An eventually constant sequence satisfies a linear recurrence, so the generating function $\sum_n e_c^{p_0,q_0}(\cdots)s^{12n}$ is rational in s. Since the target ring $\mathbb{Z}[x,y,(xy)^{-1}]$ is discrete, the full series $\sum_n \mathrm{HD}(\cdots)s^{12n}$ lies in $\mathbb{Z}[x,y,(xy)^{-1}](s)$. ■

The explicit Euler product $E(u;s)$ built from Table 1 agrees with $Z_{\mathrm{Triv}}(u;t)$ at $u=1$ by [BPS22, Thm. 8.9] and to arbitrary dimensional precision for general u by Theorem 2.12. We conjecture exact agreement:

Conjecture 2.14. For each reduction type Θ in Table 1, let $y_\Theta(u;s)$ denote the argument of $\zeta_{\mathbb{P}^1}$ in the displayed Euler factor, and set

$$E(u;s) := u^2 \cdot \mathbb{L} \cdot \prod_\Theta \zeta_{\mathbb{P}^1}\left(y_\Theta(u;s)\right). \tag{26}$$

Then $[u^m]Z_{\mathrm{Triv}}(u;t) = [u^m]E(u;s)$ in $\widehat{\mathcal{M}}_{\mathbb{L}}(s)$ for each $m \geq 2$.

Since $E(u;s)$ is rational in both u and s, the conjecture would give strict bivariate rationality of $Z_{\mathrm{Triv}}(u;t)$; it is possible, however, that the coefficient-wise rationality of Theorem 2.13 is optimal, since the stabilization threshold $n_0(m,N)$ grows with m and no uniform bound is available. The obstruction is the dimension deficiency $10n+2$ versus $12n$, which prevents jet evaluation surjectivity when the multiplicative sector carries more than $\frac{1}{3}$ of the discriminant. As an illustration of the small-height obstruction: the configuration $\mathrm{I}_7 + \mathrm{I}_3 + 2\mathrm{I}_1$ at height $n = 1$ satisfies the Shioda–Tate bound $T(S) = 10 \leq \rho(S)$ but does not appear in Persson's classification [Per90], since the discriminant map cannot realize the root multiplicities $(7,3,1,1)$ within $H^0(\mathcal{O}(4)) \oplus H^0(\mathcal{O}(6))$. For $n \geq 4$, the jet evaluation map is surjective and this pattern becomes realizable.

Over $k = \mathbb{F}_q$ with $\mathrm{char}(\mathbb{F}_q) \neq 2,3$, the dimensional error bounds of Theorem 2.12 translate via the Lang–Weil estimates [LW54] into numerical bounds: for each m and N, the normalized point counting error satisfies $|\#_q(\mathcal{W}_n^{\min}(m))\cdot q^{-10n} - R^{\#}_{m,N}\big|_{s^{12n}}| \leq C_{m,N}\cdot q^{-(N+1)}$ for all n sufficiently large, where $R^{\#}_{m,N} \in \mathbb{Q}(s)$ is the point counting specialization of $R_{m,N}$. This gives approximate weighted point counting rationality, parallel to the motivic statement; strict rationality would require showing that $H^i_{\acute{e}t,c}(\mathcal{W}_n^{\min}(m)_{/\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ stabilizes weight-by-weight as $n \to \infty$. The weight bounds for stacks with affine stabilizers established by Sun [Sun12] provide the necessary framework, but the stabilization itself is open. In particular, the normalized weighted point counts $\#_q(\mathcal{W}_n^{\min}(m)) \cdot q^{-10n}$ are approximable to arbitrary q-adic precision by rational functions of s, but strict rationality of the generating function $\sum_n \#_q(\mathcal{W}_n^{\min}(m))s^{12n} \in \mathbb{Q}[\![s]\!]$ remains open.

The codimension-2 class $\mathbb{L}^2 - \mathbb{L}$ from Lemma 2.10, computed on the coprime locus $\{(a_4(v), a_6(v)) \neq (0,0)\}$, combines with the non-coprime contribution $\mathbb{L}^2$ to give the full 1-jet class $2\mathbb{L}^2 - \mathbb{L}$ of $\{\mathrm{ord}_v(\Delta) \geq 2\}$ in $\mathbb{A}^4$. Under point counting at a closed point v of degree i, this specializes to $2|v|^2 - |v|$, reproducing the local computation in de Jong [dJ02, Rem. 4.22], who counts pairs (A,B) in $\mathcal{O}_{2v} \cong \kappa(v)[\epsilon]/(\epsilon^2)$ satisfying $4\alpha^3 + 27\beta^2 \equiv 0 \pmod{v^2}$. Poonen [Poo03, Thm. 3.4 and subsequent remark] assembles these local factors into an infinite Euler product over closed points of $\mathbb{P}^1$: the density of $(a_4, a_6) \in V_n(\mathbb{F}_q)$ with $\Delta = 4a_4^3 + 27a_6^2$ squarefree converges, as $n \to \infty$, to

$$\gamma_q = \frac{q^3}{(q-1)^2(q+1)} \prod_{v \in \mathbb{P}^1_{\mathbb{F}_q}} \left(1 - \frac{2|v|^2 - |v|}{|v|^4}\right),$$

resolving de Jong's question in [dJ02, Rem. 4.22]. The motivic local factor $\mathbb{L}^2 - \mathbb{L}$ of Lemma 2.10 lifts the coprime part of de Jong's computation to $K_0(\mathrm{Var}_k)$. The motivic limit $\zeta_{\mathbb{P}^1 \smallsetminus V_\sigma}(-\alpha)$ of Proposition 2.11(3), with $\alpha = \mathbb{L}^{-3}(\mathbb{L}-1)$, specializes under point counting (with $\mathbb{L} \mapsto q$) to Poonen's infinite arithmetic Euler product; the conjectural finite Euler product of Conjecture 2.14 would package this into finitely many Kapranov factors via $\zeta_{\mathbb{P}^1}(y) = \prod_{v \in \mathbb{P}^1}(1 - y^{\deg v})^{-1}$.

The approximate rationality of Theorem 2.12 is the natural analog, for the u-refined height zeta function, of the motivic stabilization of Hurwitz spaces proved

by Landesman–Vakil–Wood [LVW24]: for degree $d \leq 5$ covers of $\mathbb{P}^1$, the normalized class $\{\mathrm{Hur}_{d,g,k}\}/\mathbb{L}^{\dim}$ stabilizes in $\widehat{\mathcal{M}}_{\mathbb{L}}$ as $g \to \infty$, with error of codimension growing linearly in g (at rate g/κ_d for explicit constants $\kappa_3 = 4$, $\kappa_4 = 12$, $\kappa_5 = 40$). Strict equality (as opposed to equality modulo high codimension) is open even in the Hurwitz setting for $d \geq 3$; see [LVW24, Conj. 1.12] for the degree-3 case, which is the motivic analog of Roberts' secondary term conjecture. In our setting, the height n plays the role of g, the Vakil–Wood sieve [VW15] (via Proposition 2.11) gives codimension growth linear in n, and strict motivic rationality $[u^m]Z_{\mathrm{Triv}} \in \widehat{\mathcal{M}}_{\mathbb{L}}(s)$ remains open for the same structural reason: the inclusion–exclusion converges in the dimensional completion but need not stabilize at finite depth. Over $k = \mathbb{C}$, however, the Hodge–Deligne realization resolves this gap (Theorem 2.13).

Remark 2.15. The argument extends to $K = k(C)$ for any smooth projective curve C with $C(k) \neq \varnothing$. The Kapranov zeta function $\zeta_C(y)$ is rational by the Abel–Jacobi argument, the motivic factorization of Lemma 2.7 carries over with $\mathbb{P}^1$ replaced by C, and the Vakil–Wood stabilization (Proposition 2.11) applies with $\mathrm{Sym}^\ell(\mathbb{P}^1 \smallsetminus V_\sigma)$ replaced by $\mathrm{Sym}^\ell(C \smallsetminus V_\sigma)$. For $g(C) \geq 1$, the Hermite interpolation map on $H^0(\mathcal{O}_C(6n))$ may fail to be surjective, producing rational (not polynomial) corrections on the isotrivial loci. The approximate rationality conclusion $[u^m]Z_{\mathrm{Triv},C}(u;t) \in \overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}$ continues to hold for each m.

Remark 2.16. Assume $\mathrm{char}(k) \neq 2, 3$. The local factor coefficients A_Θ in Table 1 arise from two distinct mechanisms (cf. [BPS22, Thm. 1.6])

Additive types. For each additive Kodaira type $\Theta \in \{\mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*(j \neq 0, 1728), \mathrm{I}_0^*(j \in \{0, 1728\}), \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*\}$, the coefficient A_Θ is the normalized one-fiber motivic class of the stratum $\mathcal{W}^\Theta_{n,\mathbb{P}^1}$ parametrizing minimal fibrations with exactly one singular fiber of type Θ and semistable fibers elsewhere. After dividing by $\mathrm{Aut}(\mathbb{P}^1) = \mathrm{PGL}_2$[1] and the height-dependent factor $\mathbb{L}^{10n-20}$:

$$A_\Theta := \frac{\{\mathcal{W}^\Theta_{n,\mathbb{P}^1}\}}{\{\mathrm{PGL}_2\}\,\mathbb{L}^{10n-20}}.$$

This normalization differs from that in the proof of Lemma 2.7 ($A_j = \{\mathcal{W}^\Theta_{n,\mathbb{P}^1}\}/(\{\mathbb{P}^1\} \cdot \mathbb{L}^{10n})$) by a universal factor independent of Θ, absorbed into the prefactor.

Cusp families. For the multiplicative family I_k ($k \geq 1$), no additive vanishing is imposed on (a_4, a_6): the contact order $k = \mathrm{ord}_v(\Delta)$ is determined entirely by the nonlinear discriminant $\Delta = 4a_4^3 + 27a_6^2$. The coefficient A_{I_k} is therefore *not* the motivic class of a "one-I_k-fiber stratum": the number of multiplicative fibers is emergent from the Weierstrass data, not prescribed. Rather, $A_{\mathrm{I}_k} = \mathbb{L}^{18}$ for all $k \geq 1$ is the normalized local multiplicative class from the single-point jet computation of Definition 2.5 (originally [HP19, Cor. 2]). For the additive cusp family I_k^* ($k \geq 1$), the local condition imposes additive vanishing $(p, q) = (2, 3)$ (the same as $\mathrm{I}_0^*(j \neq 0, 1728)$; cf. [BPS22, Thm. 1.6]) together with a cusp contact order k determined by the discriminant; the normalized class is $A_{\mathrm{I}_k^*} = \mathbb{L}^{14} - \mathbb{L}^{13}$ for all $k \geq 1$.

[1] The *unparameterized* $\mathbb{P}^1_k$ corresponds to taking the PGL_2 stack quotient; motivically this factors out $\{\mathrm{PGL}_2\} = \mathbb{L}(\mathbb{L}^2 - 1)$. See [PS25] for a comprehensive treatment.

For I_k^*, the k-independence reflects the fact that the additive vanishing condition $(p,q)=(2,3)$ is the same for all k; the contact order k is emergent from the residual discriminant. For I_k, there is no per-k local factor: the coefficient $\mathbb{L}^{18}$ is the fiber class of the map from coprime Weierstrass data to the discriminant divisor, and individual I_k fibers emerge from the root structure of $\Delta = 4a_4^3 + 27a_6^2$. In both cases, the k-independence is what allows the cusp resummation (Lemma 2.4) to collapse the k-dependence into $(1-us)^{-1}$.

Then [BPS22, Thm. 1.8] determines the following normalized one-fiber motivic classes.

Conventions. In Tables 1–4, $\delta(s) := 1-us$ denotes the cusp resummation denominator of Lemma 2.4; this is not to be confused with the Weierstrass discriminant $\Delta = 4a_4^3+27a_6^2$. For each reduction type Θ, let $y_\Theta(u;s)$ denote the local monomial in the displayed denominator; we record only the reduced factor $(1-y_\Theta)^{-1}$, the full $\mathbb{P}^1$-contribution being

$$(1-y_\Theta)^{-\{\mathbb{P}^1\}} = \frac{1}{\big(1-y_\Theta\big)\big(1-\mathbb{L}\,y_\Theta\big)}.$$

3. Applications to modular curves with prescribed level structure

We apply the Main Theorem to the genus-0 modular curves $\overline{\mathcal{M}}_1(N)$ parametrizing generalized elliptic curves with level-N structure $\Gamma_1(N)$, introduced by [DR73] (see also [Con07, §2]). The fine modular curve $\overline{\mathcal{M}}_1(N)$ parametrizes families $(E,S,P)\to B$ where $(E,S)\to B$ is a semistable elliptic curve with section S and $P\in E^{\mathrm{sm}}[N](B)$ is an N-torsion section such that the divisor $P+S$ is relatively ample [KM85, §1.4]. We focus on $N=2,3,4$, where the modular curves are genuinely stacky. Throughout, let k be a perfect field with $\mathrm{char}(k)\neq 2,3$.

Corollary 3.1. *Let k be a perfect field with $\mathrm{char}(k)\neq 2,3$, set $K=k(t)$, and let $s=t^{1/12}$. For $N\in\{2,3,4\}$, the trivial-lattice-rank-weighted motivic height zeta function*

$$Z_{\mathrm{Triv}}^{\Gamma_1(N)}(u;t) := \sum_{n\geq 0}\sum_{T\geq 2} u^T\left\{\mathcal{W}_n^{\Gamma_1(N)}(T)\right\}t^n$$

satisfies $[u^m]Z_{\mathrm{Triv}}^{\Gamma_1(N)} \in \overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}$ for each $m\geq 2$. More precisely, Theorem 2.12 applies with $\overline{\mathcal{M}}_{1,1}$ replaced by $\overline{\mathcal{M}}_1(N)\cong\mathcal{P}(w_1,w_2)$ and the local factor types J, coefficients A_Θ, and Euler factors as given in Tables 2–4 for $(w_1,w_2)=(2,4)$, $(1,3)$, $(1,2)$ respectively.

Proof. The proof of Theorem 2.12 uses only the formal properties of the height–moduli framework of [BPS22] applied to a proper polarized cyclotomic stack $\mathcal{P}(w_1,w_2)$, and each ingredient applies to $\mathcal{P}(2,4)$, $\mathcal{P}(1,3)$, and $\mathcal{P}(1,2)$ with the admissible Kodaira types restricted by the level structure, the surjectivity threshold adjusted to $\dim H^0(\mathcal{O}(w_1 n))$ and $\dim H^0(\mathcal{O}(w_2 n))$, and $\mathbb{L}^{10n}$ replaced by $\mathbb{L}^{(w_1+w_2)n}$.

The only ingredient requiring case-by-case verification is the codimension-2 property of extra multiple roots (Lemma 2.10), since the discriminant of each $\mathcal{P}(w_1,w_2)$ has a different functional form. For $\mathcal{P}(4,6)$, the discriminant $\Delta = 4a_4^3+27a_6^2$ is irreducible and the computation is in Lemma 2.10. For $\mathcal{P}(2,4)$, the discriminant

Reduction type Θ	$\gamma : (\nu(a_4),\ \nu(a_6))$	(r,a)	$m_\nu - 1$	$\mathbb{P}^1$-Euler factor in $Z_{\mathrm{Triv}}(u;t)$
$\mathrm{I}_{k\geq 1}$ $(j=\infty)$	$(0,0)$	$\emptyset$	$k-1$	$\dfrac{1}{1-\mathbb{L}^{18}\,s\,\delta(s)^{-1}}$
II $(j=0)$	$(\geq 1,\ 1)$	$(6,1)$	0	$\dfrac{1}{1-\mathbb{L}^{17}\,s^2}$
III $(j=1728)$	$(1,\ \geq 2)$	$(4,1)$	1	$\dfrac{1}{1-\mathbb{L}^{16}\,u\,s^3}$
IV $(j=0)$	$(\geq 2,\ 2)$	$(3,1)$	2	$\dfrac{1}{1-\mathbb{L}^{15}\,u^2\,s^4}$
$\mathrm{I}^*_{k\geq 1}$ $(j=\infty)$	$(2,\ 3)$	$(2,1)$	$k+4$	$\dfrac{1}{1-(\mathbb{L}^{14}-\mathbb{L}^{13})\,u^5\,s^7\,\delta(s)^{-1}}$
I^*_0 $(j\neq 0,1728)$	$(2,\ 3)$	$(2,1)$	4	$\dfrac{1}{1-(\mathbb{L}^{14}-\mathbb{L}^{13})\,u^4\,s^6}$
I^*_0 $(j=0)$	$(\geq 3,\ 3)$	$(2,1)$	4	$\dfrac{1}{1-\mathbb{L}^{13}\,u^4\,s^6}$
I^*_0 $(j=1728)$	$(2,\ \geq 4)$	$(2,1)$	4	$\dfrac{1}{1-\mathbb{L}^{13}\,u^4\,s^6}$
IV^* $(j=0)$	$(\geq 3,\ 4)$	$(3,2)$	6	$\dfrac{1}{1-\mathbb{L}^{12}\,u^6\,s^8}$
III^* $(j=1728)$	$(3,\ \geq 5)$	$(4,3)$	7	$\dfrac{1}{1-\mathbb{L}^{11}\,u^7\,s^9}$
II^* $(j=0)$	$(\geq 4,\ 5)$	$(6,5)$	8	$\dfrac{1}{1-\mathbb{L}^{10}\,u^8\,s^{10}}$

TABLE 1. Local factors for $\mathcal{P}(4,6)$.

$\Delta = 16\,a_4^2(a_2^2-4a_4)$ factors. The locus $\{a_4(v)=0,\ a_2(v)\neq 0\}$ gives $\mathrm{ord}_v(\Delta)\geq 2$ at codimension 1, but corresponds to a type III fiber ($\gamma=(\geq 1,1)$ in Table 2), already captured by the additive Euler product. On the residual coprime locus $\{a_4(v)\neq 0\}$, $\mathrm{ord}_v(\Delta)=\mathrm{ord}_v(a_2^2-4a_4)$, and the codimension-2 property holds: $a_2(v)^2-4a_4(v)=0$ cuts codimension 1, and the derivative condition $2a_2(v)\,a_2'(v)-4a_4'(v)=0$ is nontrivial (since $a_2(v)\neq 0$ when $a_4(v)\neq 0$ and $a_2(v)^2=4a_4(v)$), cutting a further codimension 1. For $\mathcal{P}(1,3)$, the discriminant $\Delta = a_3^3(a_1^3-27a_3)$ factors similarly.

The codimension-1 locus $\{a_3(v)=0,\ a_1(v)\neq 0\}$ produces multiplicative fibers I_{3k} governed by Hermite interpolation on $H^0(\mathcal{O}(3n))$, not by the discriminant stabilization; on the residual locus $\{a_3(v)\neq 0\}$, the codimension-2 property for $a_1^3-27a_3$ follows as above (the derivative condition $3a_1^2\,a_1'-27a_3'=0$ is nontrivial since the coefficient of a_3' is $-27\neq 0$). For $\mathcal{P}(1,2)$ the verification is analogous; the only additive type is I_0^* (Table 4). ■

3.1. **Level-2 structure.** We consider the height moduli

$$\mathcal{W}_n^{\Gamma_1(2)}\big(\mathcal{P}(2,4),\mathcal{O}(1)\big) \;=\; \mathcal{M}_{n,\mathbb{P}^1_k}\big(\overline{\mathcal{M}}_1(2),\mathcal{L}\big),$$

where $\overline{\mathcal{M}}_1(2)\cong\mathcal{P}(2,4)$ over $\mathbb{Z}\big[\frac{1}{2}\big]$ is the moduli stack of generalized elliptic curves with $\Gamma_1(2)$-structure (cf. [Beh06, §1.3]). Equivalently, $\overline{\mathcal{M}}_1(2)$ admits the universal Weierstrass presentation

$$y^2=x^3+a_2x^2+a_4x \qquad \text{with} \qquad (a_2,a_4)\in H^0\big(\mathbb{P}^1,\mathcal{O}(2n)\big)\times H^0\big(\mathbb{P}^1,\mathcal{O}(4n)\big).$$

Reduction type Θ	$\gamma:(\nu(a_2),\ \nu(a_4))$	(r,a)	m_v-1	$\mathbb{P}^1$-Euler factor in $Z_{\mathrm{Triv}}^{\Gamma_1(2)}(u;t)$
$\mathrm{I}_{k\geq 1}$ $(j=\infty)$	$(0,\,0)$	$\emptyset$	$k-1$	$\dfrac{1}{1-\mathbb{L}^{10}s\,\delta(s)^{-1}}$
III $(j=1728)$	$(\geq 1,\,1)$	$(4,1)$	1	$\dfrac{1}{1-\mathbb{L}^{9}u\,s^3}$
$\mathrm{I}^*_{k\geq 1}$ $(j=\infty)$	$(1,\,2)$	$(2,1)$	$k+4$	$\dfrac{1}{1-(\mathbb{L}^8-\mathbb{L}^7)u^5s^7\,\delta(s)^{-1}}$
I_0^* $(j\neq 0,1728)$	$(1,\,2)$	$(2,1)$	4	$\dfrac{1}{1-(\mathbb{L}^8-\mathbb{L}^7)u^4s^6}$
I_0^* $(j=0)$	$(\geq 2,\,2)$	$(2,1)$	4	$\dfrac{1}{1-\mathbb{L}^7u^4s^6}$
I_0^* $(j=1728)$	$(1,\,\geq 3)$	$(2,1)$	4	$\dfrac{1}{1-\mathbb{L}^7u^4s^6}$
III^* $(j=1728)$	$(\geq 2,\,3)$	$(4,3)$	7	$\dfrac{1}{1-\mathbb{L}^6u^7s^9}$

TABLE 2. Local factors for $\mathcal{P}(2,4)$.

3.2. **Level-3 structure.** We consider the height moduli

$$\mathcal{W}_n^{\Gamma_1(3)}\big(\mathcal{P}(1,3),\mathcal{O}(1)\big) \;=\; \mathcal{M}_{n,\mathbb{P}^1_k}\big(\overline{\mathcal{M}}_1(3),\mathcal{L}\big),$$

where $\overline{\mathcal{M}}_1(3) \cong \mathcal{P}(1,3)$ over $\mathbb{Z}\left[\frac{1}{3}\right]$ is the moduli stack of generalized elliptic curves with $\Gamma_1(3)$-structure (cf. [HM17, Prop. 4.5]). Equivalently, $\overline{\mathcal{M}}_1(3)$ admits the universal Weierstrass presentation

$$y^2 + a_1 xy + a_3 y = x^3 \qquad \text{with} \qquad (a_1, a_3) \in H^0\big(\mathbb{P}^1, \mathcal{O}(n)\big) \times H^0\big(\mathbb{P}^1, \mathcal{O}(3n)\big).$$

Reduction type Θ	$\gamma : (\nu(a_1),\ \nu(a_3))$	(r,a)	$m_\nu - 1$	$\mathbb{P}^1$-Euler factor in $Z^{\Gamma_1(3)}_{\mathrm{Triv}}(u;t)$
$\mathrm{I}_{k\geq 1}$ $(j=\infty)$	$(0,\ 0)$	$\emptyset$	$k-1$	$\dfrac{1}{1-\mathbb{L}^6 s\,\delta(s)^{-1}}$
IV $(j=0)$	$(\geq 1,\ 1)$	$(3,1)$	2	$\dfrac{1}{1-\mathbb{L}^5 u^2 s^4}$
IV* $(j=0)$	$(\geq 1,\ 2)$	$(3,2)$	6	$\dfrac{1}{1-\mathbb{L}^4 u^6 s^8}$

TABLE 3. Local factors for $\mathcal{P}(1,3)$.

3.3. **Level-4 structure.** We consider the height moduli

$$\mathcal{W}_n^{\Gamma_1(4)}\big(\mathcal{P}(1,2), \mathcal{O}(1)\big) \;=\; \mathcal{M}_{n,\mathbb{P}^1_k}\big(\overline{\mathcal{M}}_1(4), \mathcal{L}\big),$$

where $\overline{\mathcal{M}}_1(4) \cong \mathcal{P}(1,2)$ over $\mathbb{Z}\left[\frac{1}{2}\right]$ is the moduli stack of generalized elliptic curves with $\Gamma_1(4)$-structure (cf. [Mei22, Ex. 2.1]). Equivalently, $\overline{\mathcal{M}}_1(4)$ admits the universal Weierstrass presentation

$$y^2 + a_1 xy + a_1 a_2 y = x^3 + a_2 x^2 \qquad \text{with} \qquad (a_1, a_2) \in H^0\big(\mathbb{P}^1, \mathcal{O}(n)\big) \times H^0\big(\mathbb{P}^1, \mathcal{O}(2n)\big).$$

Reduction type Θ	$\gamma : (\nu(a_1),\ \nu(a_2))$	(r,a)	$m_\nu - 1$	$\mathbb{P}^1$-Euler factor in $Z^{\Gamma_1(4)}_{\mathrm{Triv}}(u;t)$
$\mathrm{I}_{k\geq 1}$ $(j=\infty)$	$(0,\ 0)$	$\emptyset$	$k-1$	$\dfrac{1}{1-\mathbb{L}^4 s\,\delta(s)^{-1}}$
I_0^* $(j=0)$	$(\geq 1,\ 1)$	$(2,1)$	4	$\dfrac{1}{1-\mathbb{L}^3 u^4 s^6}$

TABLE 4. Local factors for $\mathcal{P}(1,2)$.

The level-structure framework applies uniformly to all genus-0 modular curves $\overline{\mathcal{M}}_1(N) \cong \mathcal{P}(w_1, w_2)$, yielding strict rationality of the full torsion-weighted motivic height zeta function.

Proposition 3.2. *Let k be a perfect field with* $\mathrm{char}(k) \neq 2,3$*, and set $K = k(t)$. The torsion-weighted motivic height zeta function*

$$Z_{\mathrm{tor}}(\tau;t) \;:=\; \sum_{n\geq 0} \sum_{N\geq 1} \tau^N \big\{\mathcal{W}_n^{N\text{-}tor}\big\}\, t^n$$

is a polynomial in τ of degree at most 12, with coefficients that are rational functions of t in $K_0(\mathrm{Stck}_k)[\mathbb{L}^{-1}](t)$. The cases $N \in \{2,3,4\}$ are computed explicitly in Corollary 3.1 and Tables 2–4.

Proof. A non-isotrivial E/K with a K-rational point of exact order N gives a non-constant rational map $\mathbb{P}^1_k \dashrightarrow \overline{\mathcal{M}}_1(N)$ by [BPS22, Thm. 3.3 & Prop. 5.8], which exists only when $g(X_1(N)) = 0$, i.e. $N \in \{1,2,3,4,5,6,7,8,9,10,12\}$ (cf. [KM85, §1.3]). For each such N, the identification $\overline{\mathcal{M}}_1(N) \cong \mathcal{P}(w_1, w_2)$ places the height moduli in the BPS framework, and [BPS22, Thm. 8.9] gives rationality of $\sum_n \{\mathcal{W}_n^{\Gamma_1(N)}\} t^n$. The forgetful map $\mathcal{W}_n^{\Gamma_1(N)} \to \mathcal{W}_n^{N\text{-tor}}$ is representable with finite fibers, so the two have the same rationality properties after inverting $\mathbb{L}$. Since the sum over N is finite, $Z_{\mathrm{tor}}(\tau; t)$ is rational. ∎

4. Irrationality of the Néron–Severi and Mordell–Weil specializations

The approximate rationality of $Z_{\mathrm{Triv}}(u; t)$ rests on the fact that the trivial lattice rank $T(S)$ is governed by *local* reduction data: it depends only on the multiset of fiber component numbers m_v, hence is constant on each Kodaira stratum $\mathcal{W}_n^{\min,(\mathfrak{f})}$ in the finite constructible stratification. This locality is what makes the evaluation-map factorization, the isotrivial Euler products, and the motivic discriminant stabilization work: unordered collections of local factors assemble into a finite Euler product, and the residual multiplicative sector stabilizes in the dimensional completion. Over $k = \mathbb{C}$, the weight bound in mixed Hodge theory then promotes approximate motivic rationality to strict Hodge–Deligne rationality (Theorem 2.13).

In contrast, the Mordell–Weil rank $\mathrm{rk}(E/K)$ is not determined by the fiber configuration. Even on a fixed Kodaira stratum $\mathcal{W}_n^{\min,(\mathfrak{f})}$, the rank typically varies, reflecting genuinely *global* constraints rather than local reduction data. Since $T(S)$ is constant on $\mathcal{W}_n^{\min,(\mathfrak{f})}$, the Shioda–Tate formula (1) shows that variation of $\mathrm{rk}(E/K)$ is equivalent to variation of the Picard rank $\rho(S)$. Inside $\mathcal{W}_n^{\min,(\mathfrak{f})}$, imposing that $\mathrm{NS}(S_{\bar{k}})$ contain additional algebraic classes independent of the trivial lattice — equivalently, that $\rho(S)$, hence $\mathrm{rk}(E/K)$, jump — is an algebraic condition. Over $\mathbb{C}$, these conditions are modeled by Noether–Lefschetz loci for the variation of Hodge structure on H^2, and the theorem of Cattani–Deligne–Kaplan [CDK95] shows that such loci are, in general, only a *countable union of closed algebraic subsets*. In particular, unlike the Kodaira stratification at fixed height n (which is finite by Proposition 2.1), refinements by Néron–Severi *lattice type* are not expected to admit a finite constructible stratification.

This suggests that the local-to-global factorization mechanism producing a finite Euler product for Z_{Triv} should structurally fail for the Mordell–Weil and Néron–Severi specializations, and motivates the following conjecture.

Conjecture 4.1. Let $k = \mathbb{C}$ and $K = \mathbb{C}(t)$, and put $s = t^{1/12}$. The specializations

$$Z_{\mathrm{MW}}(v; t) := \mathcal{Z}(1, v; t), \qquad Z_{\mathrm{NS}}(w; t) := \mathcal{Z}(w, w; t)$$

are not approximately rational in s; i.e.

$$[v^r] Z_{\mathrm{MW}}(v; t) \notin \overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}, \qquad [w^r] Z_{\mathrm{NS}}(w; t) \notin \overline{\widehat{\mathcal{M}}_{\mathbb{L}}(s)}$$

for some (conjecturally all) $r \geq 1$.

Failure of Hodge-number stabilization — i.e. the existence of some (p_0, q_0) such that $n \mapsto e_c^{p_0,q_0}\big([w^r]Z_{\mathrm{NS}}\big|_{t^n} \cdot \mathbb{L}^{-10n}\big)$ does not satisfy any linear recurrence — would imply Conjecture 4.1 by the contrapositive of the argument of Theorem 2.13.

Over $\mathbb{C}(t)$, the unboundedness of Mordell–Weil ranks is itself a major open problem. Conjecture 4.1 is logically independent: even if ranks are bounded by some R (so that Z_{MW} is polynomial of degree $\leq R$ in v), the Noether–Lefschetz obstruction persists at each fixed rank, since the Picard rank-ρ locus inside $\mathcal{W}_n^{\min}$ is carved out by a countable — but not finite constructible — union of conditions at every height n.

Over $\mathbb{F}_q(t)$, rank unboundedness is known: Ulmer's classical construction [Ulm02] produces an explicit family of elliptic curves over $\mathbb{F}_p(t)$ with arbitrarily large Mordell–Weil rank.

Theorem 4.2 ([Ulm02])**.** *For each prime p and each $r \geq 0$, there exists an elliptic curve $E/\mathbb{F}_p(t)$ with $\mathrm{rk}(E/\mathbb{F}_p(t)) \geq r$. Explicitly, the Legendre curve $y^2 = x(x+1)(x+t^d)$ over $\mathbb{F}_p(t)$ has rank growing with d.*

In particular, for each $r \geq 0$ the rank-r locus in $\mathcal{W}_n^{\min}(\mathbb{F}_p)$ is nonempty for some n. By analogy with the Noether–Lefschetz obstruction over $\mathbb{C}$, these loci are not expected to admit a finite constructible stratification compatible with the Euler product mechanism of Theorem 2.12. We conjecture that this prevents rationality in s:

Conjecture 4.3. Let $\mathrm{char}(\mathbb{F}_q) > 3$, set $K = \mathbb{F}_q(t)$, and put $s = t^{1/12}$. The specializations

$$Z_{\mathrm{MW}}(v;t) = \mathcal{Z}(1, v; t), \qquad Z_{\mathrm{NS}}(w;t) = \mathcal{Z}(w, w; t),$$

viewed as weighted point counting series over $\mathbb{F}_q$, are not coefficient-wise rational in s; i.e.

$$Z_{\mathrm{MW}}(v;t) \notin \mathbb{Q}(s)[\![v]\!], \qquad Z_{\mathrm{NS}}(w;t) \notin \mathbb{Q}(s)[\![w]\!].$$

Although Noether–Lefschetz loci are defined via Hodge theory, their arithmetic content is visible over finite fields: the loci where $\rho(S)$ jumps are detected by Frobenius eigenvalues on $H^2_{\acute{e}t}(S_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ acquiring additional algebraic coincidences (cf. [Tay25] for the analogous phenomenon for K3 surfaces).

Acknowledgements

Warm thanks to Dori Bejleri, Klaus Hulek, Scott Mullane and Matthew Satriano for helpful discussions. The author is particularly grateful to Dori Bejleri for suggesting that the Euler product should be understood as asymptotically exact, and to Will Sawin for identifying key obstructions in an earlier version — including the discriminant nonlinearity and the necessity of completing the Grothendieck ring — which motivated the present approach. The author was partially supported by the ARC grant DP210103397 and the Sydney Mathematical Research Institute.

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